Liver Transplantation

When to Refer and What's New

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Slides (PDF)

Illustrations generated with Google Gemini



Disclosures

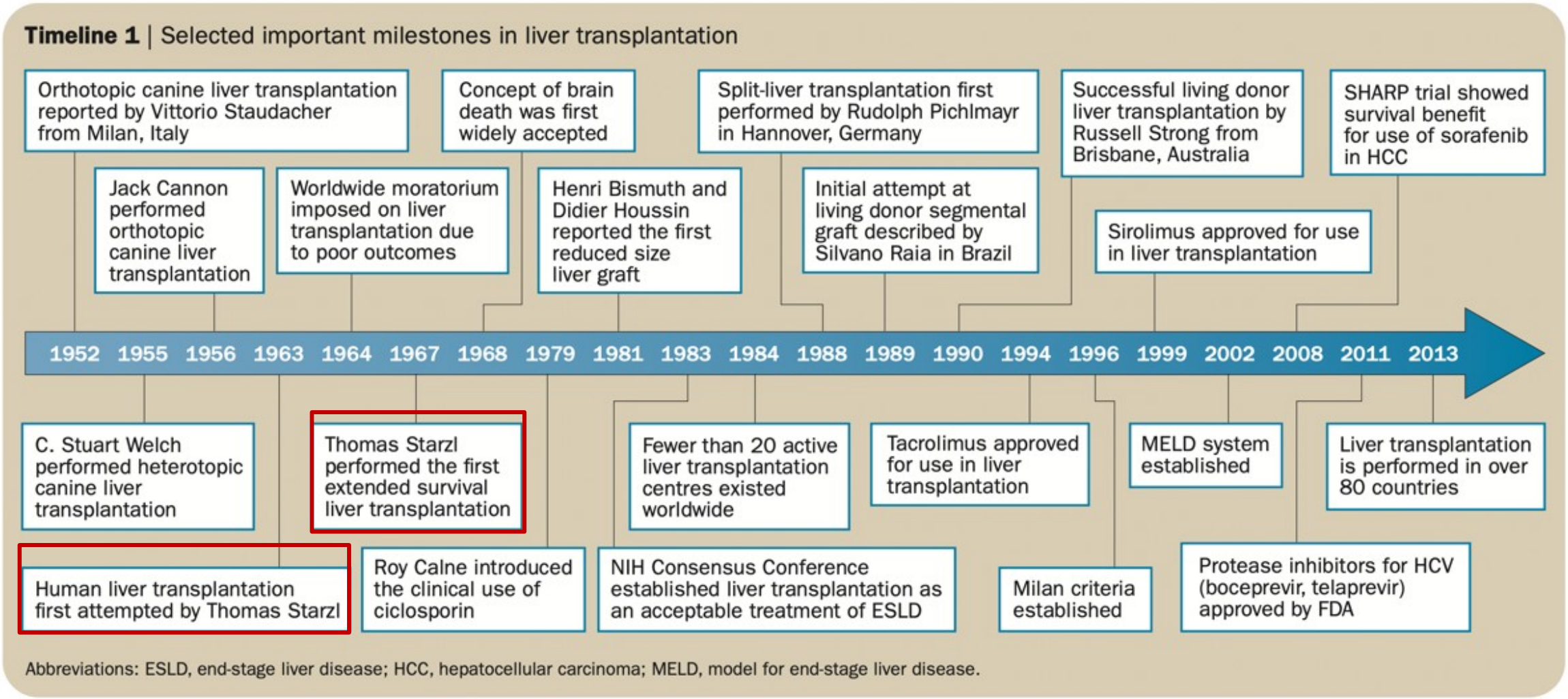
No relevant financial relationships to disclose.

Objectives

- Recognize who to refer, and who needs a call the same day
- See who can be a transplant candidate now
- Walk through what happens after the referral
- Understand where donor livers come from now
- Care for a transplant recipient on your service



How did liver transplant start?



Timeline 1 from Zarrinpar A, Busuttil RW. Nat Rev Gastroenterol Hepatol 2013;10(7):434-40 [PMID: 23752825]

Who was Thomas Starzl?


The New York Times

Dr. Thomas E. Starzl, Pioneering Liver Surgeon, Dies at 90

- 1963, Denver: first human liver transplant attempt
- 1967: first patient to live more than a year after one
- 1980: cyclosporine with steroids; survival finally good enough to offer it widely
- 1989, Pittsburgh: tacrolimus
- Died March 4, 2017, at 90

The New York Times, March 5, 2017. Starzl TE et al. Homotransplantation of the liver in humans. Surg Gynecol Obstet 1963;117:659-76 [PMID: 14100514]

How did the first liver transplants go?

SURGERY
Gynecology & Obstetrics

DECEMBER 1963
VOLUME 117
NUMBER 6

HOMOTRANSPLANTATION OF THE LIVER IN HUMANS

T. E. STARZL, M.D., F.A.C.S., T. L. MARCHIORO, M.D., K. N. VON KAULLA, M.D., G. HERMANN, M.D., R. S. BRITTAIN, M.D., and W. R. WADDELL, M.D., F.A.C.S., Denver, Colorado

- Patient 1, 1 March 1963: a 3-year-old with biliary atresia, died of bleeding on the operating table
- Patients 2 and 3 died 22 and 7½ days later, of pulmonary emboli

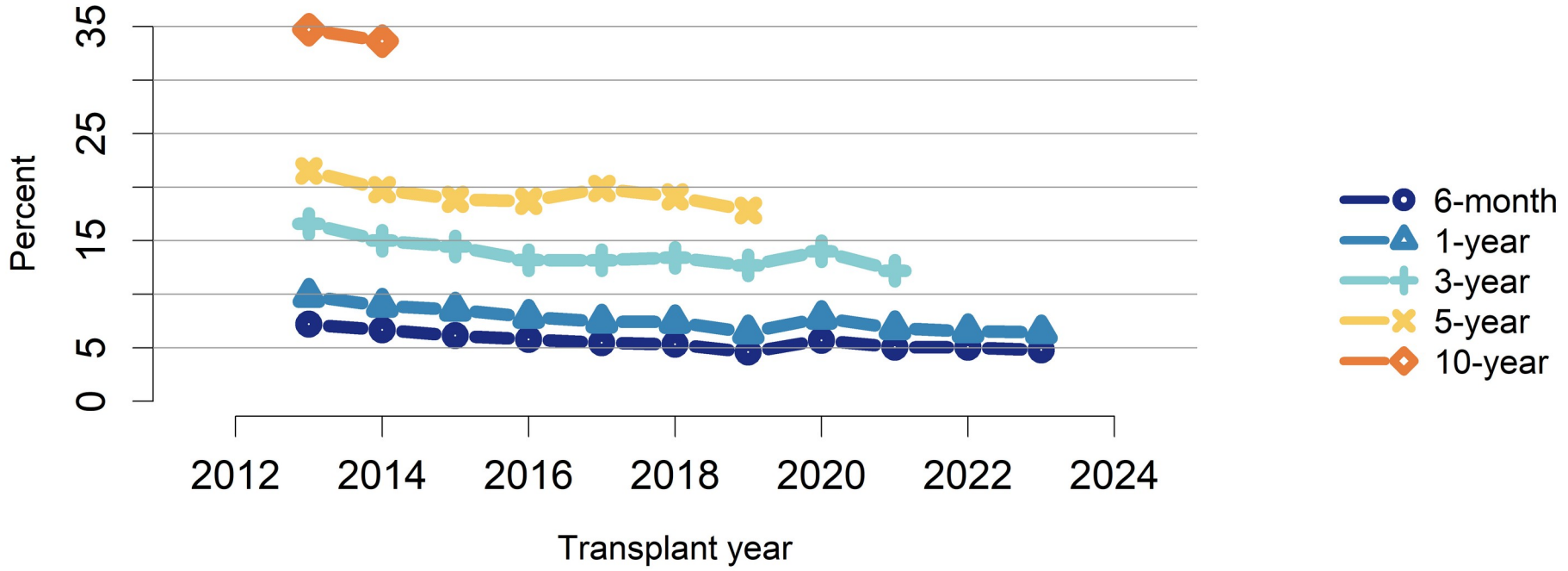
The first attempt resulted in failure at the operating table. The course of the second 2 patients establishes the feasibility of such an operation in humans, despite the fact that death occurred 22 and 7½ days after transplantation from pulmonary emboli.

Recipient patients. Patient 1 was a 3 year old white male with congenital biliary atresia (Fig. 1A). Physical development had been retarded, preoperative weight being 20 pounds. His general condition was poor, with hepatosplenomegaly, jaundice, and ascites. Total bilirubin was 20.7 milligrams per cent with a conjugated fraction of 16.7 milligrams per cent. Alkaline phosphatase was 12.8 Bodansky units. Serum glutamic-oxalacetic acid transaminase (SGOT) was 160 SF units. The hospital course prior to hepatic homotransplantation on 1 March 1963 was uneventful.

Starzl TE et al. Surg Gynecol Obstet 1963;117:659-76 [PMID: 14100514]. Scan: University of Pittsburgh D-Scholarship

How do patients do after a liver transplant?

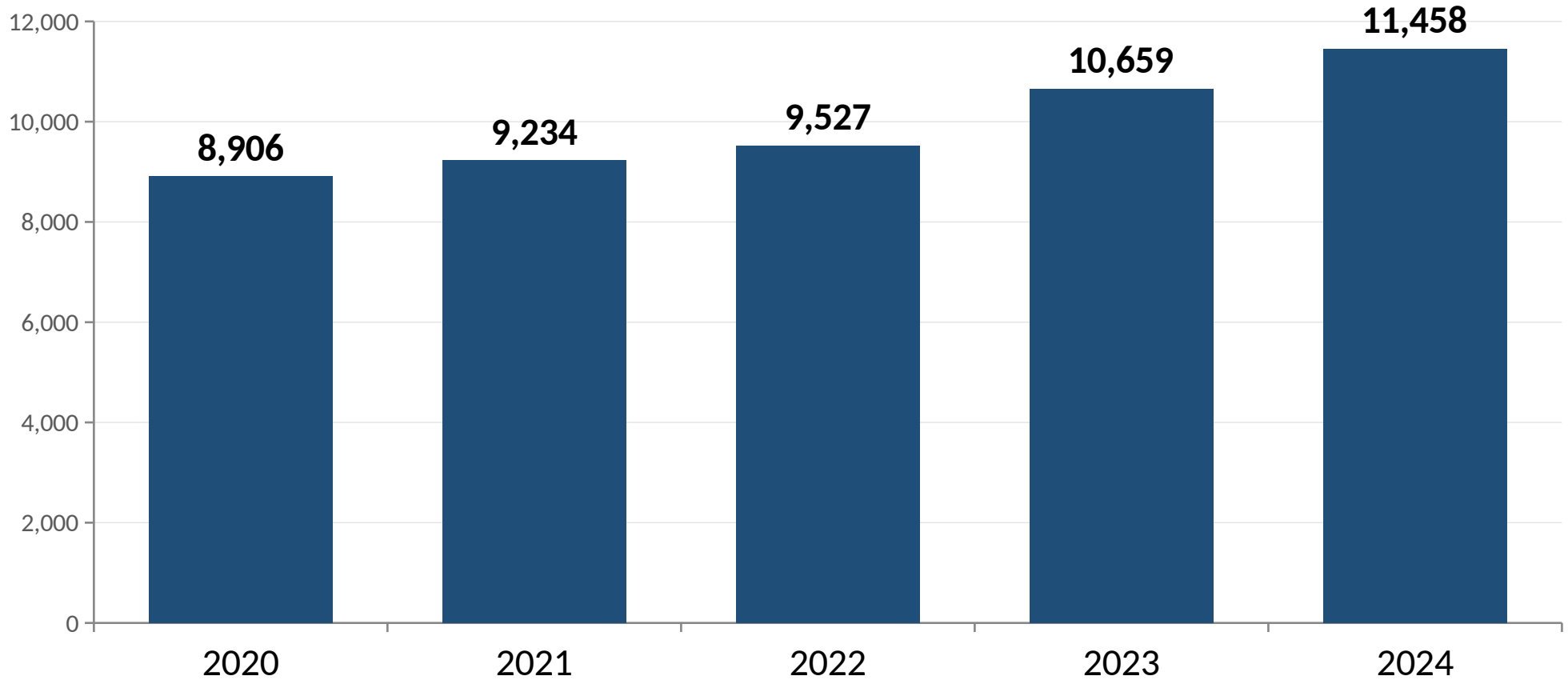
Figure LI 59: Percentages of patient deaths among adult liver transplant recipients



OPTN/SRTR 2024 Annual Data Report

OPTN/SRTR 2024 Annual Data Report, Figure LI 59. Am J Transplant 2026;26(8S1):S214-S320 [PMID: 42276697]

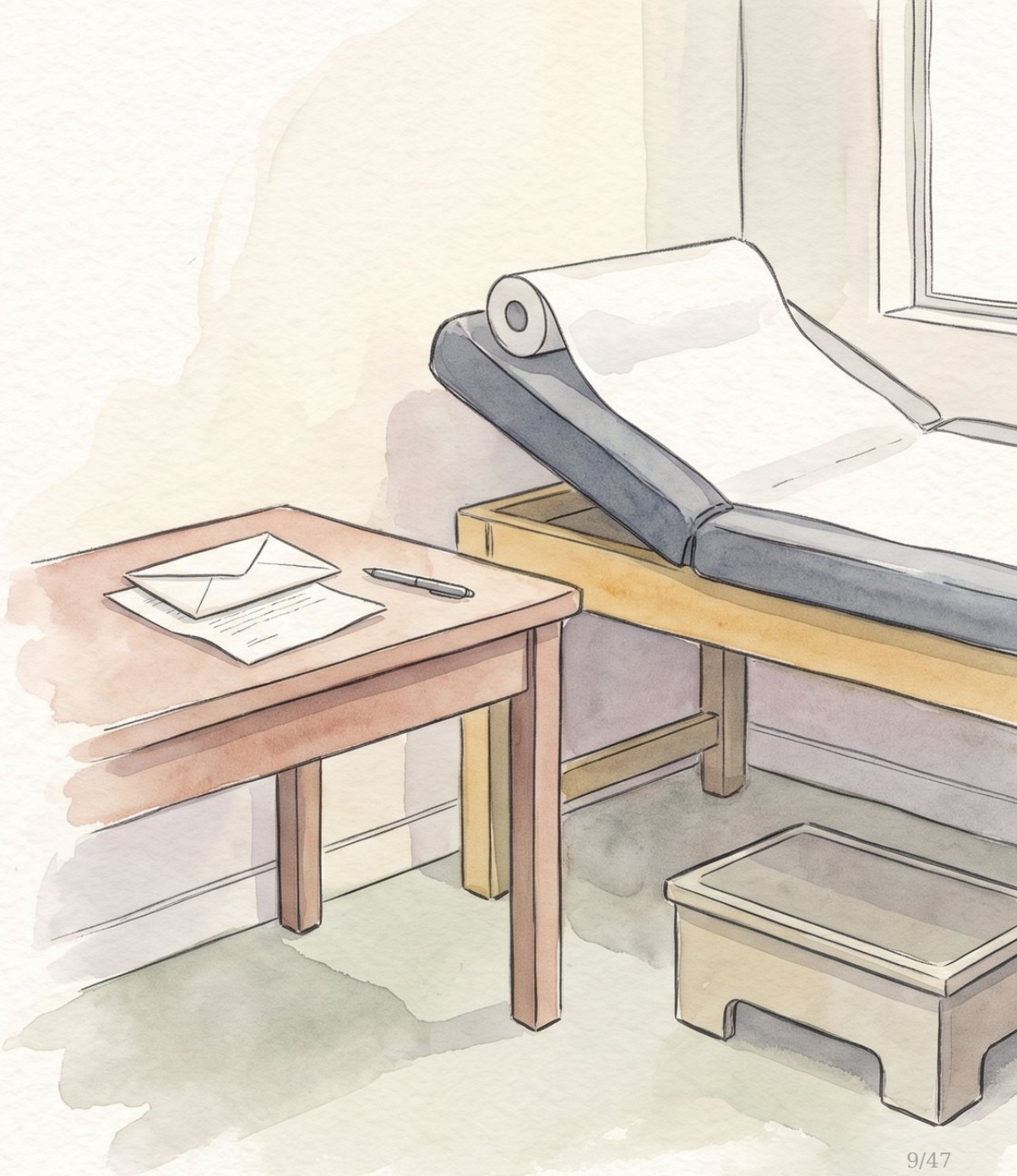
How much has liver transplant grown?



11,458 in 2024 (10,886 adults), 70% more than in 2014

OPTN/SRTR Annual Data Reports, Liver, 2020-2024 [PMIDs: 35266621, 37132348, 38431359, 39947804, 42276697]

When to refer



Where did MELD come from?

Comparative Study > Hepatology. 2000 Apr;31(4):864-71. doi: 10.1053/he.2000.5852.

A model to predict poor survival in patients undergoing transjugular intrahepatic portosystemic shunts

M Malinchoc [1], P S Kamath, F D Gordon, C J Peine, J Rank, P C ter Borg

Review > Hepatology. 2001 Feb;33(2):464-70. doi: 10.1053/jhep.2001.22172.

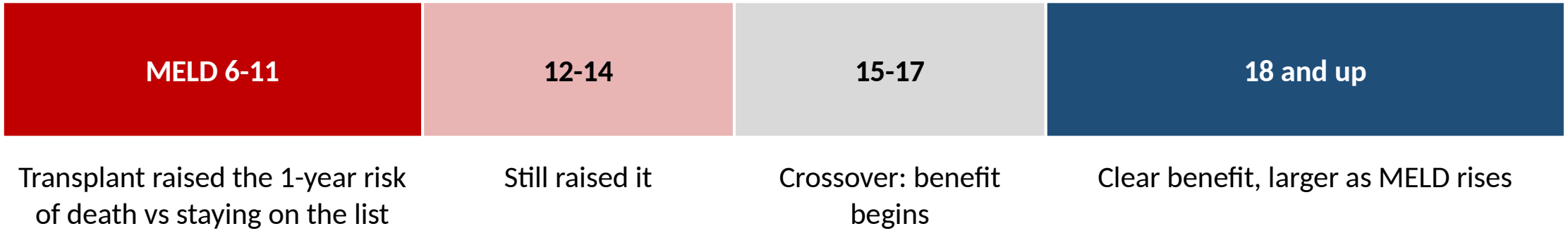
A model to predict survival in patients with end-stage liver disease

P S Kamath [1], R H Wiesner, M Malinchoc, W Kremers, T M Therneau, C L Kosberg, G D'Amico, E R Dickson, W R Kim

- 2000: built at Mayo to predict death after an elective TIPS
- 2001: worked just as well across end-stage liver disease
- 2002: became the basis for US liver allocation

Malinchoc M, Kamath PS, Gordon FD, et al. Hepatology 2000;31(4):864-71 [PMID: 10733541]. Kamath PS et al. Hepatology 2001;33(2):464-70 [PMID: 11172350]

Why did MELD 15 become the rule?



- US waitlist candidates: transplant vs staying on the list, by MELD updated on the list and just before transplant
- A 2009 decision analysis came to the same threshold of 15
- Both were before MELD-Na, MELD 3.0 and today's donor pool

Merion RM et al. Am J Transplant 2005;5(2):307-13 [PMID: 15643990]. Perkins JD et al. Liver Transpl 2009;15(2):242-54 [PMID: 19177441]

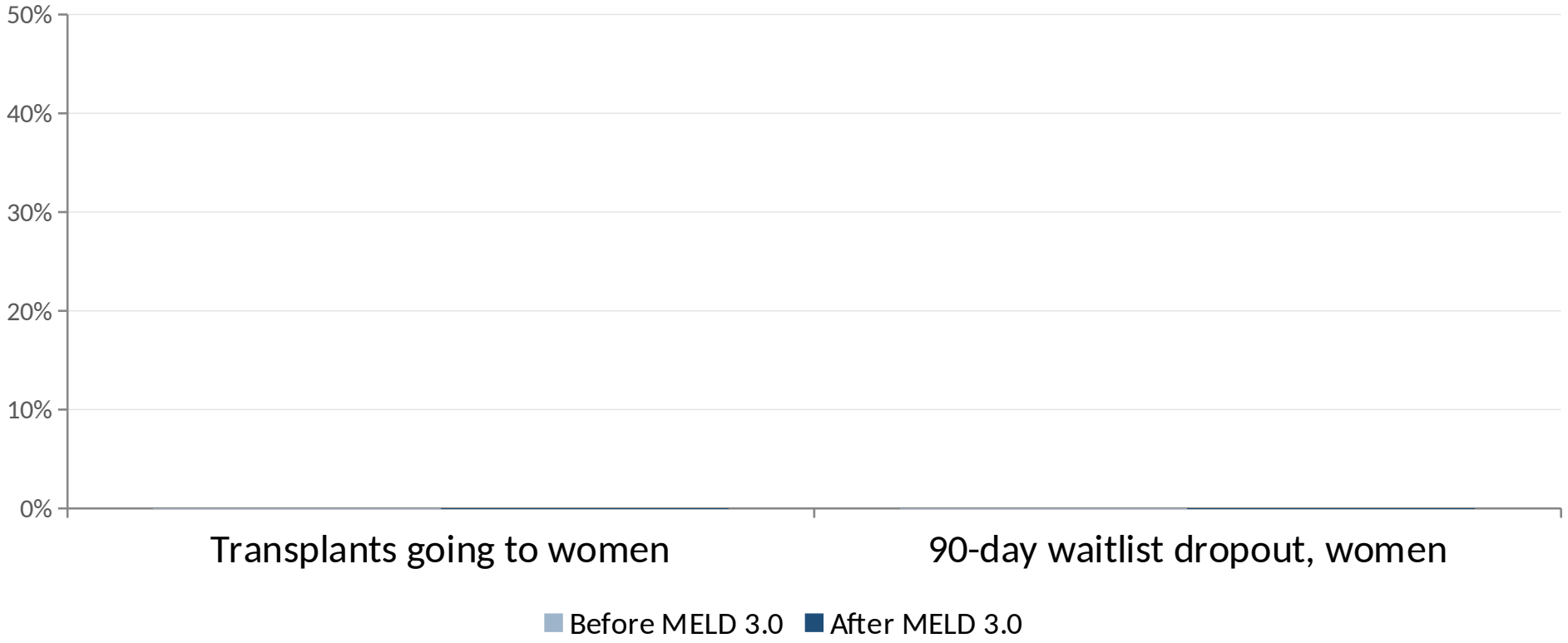
What's new in MELD 3.0?

	MELD-Na	MELD 3.0
Bilirubin	✓	✓
INR	✓	✓
Creatinine	✓ (cap 4.0)	✓ (cap 3.0)
Sodium	✓	✓
Albumin		✓
Female sex		✓ **(+1.33)**
Interaction terms	MELD × Na	Bili × Na, albumin × Cr

- Used for allocation since July 2023
- 90-day mortality: C-statistic 0.869 vs 0.862
- In simulation, at least 20 fewer waitlist deaths a year

Kim WR, Mannalithara A, Heimbach JK, et al. Gastroenterology 2021;161(6):1887-1895.e4 [PMID: 34481845]

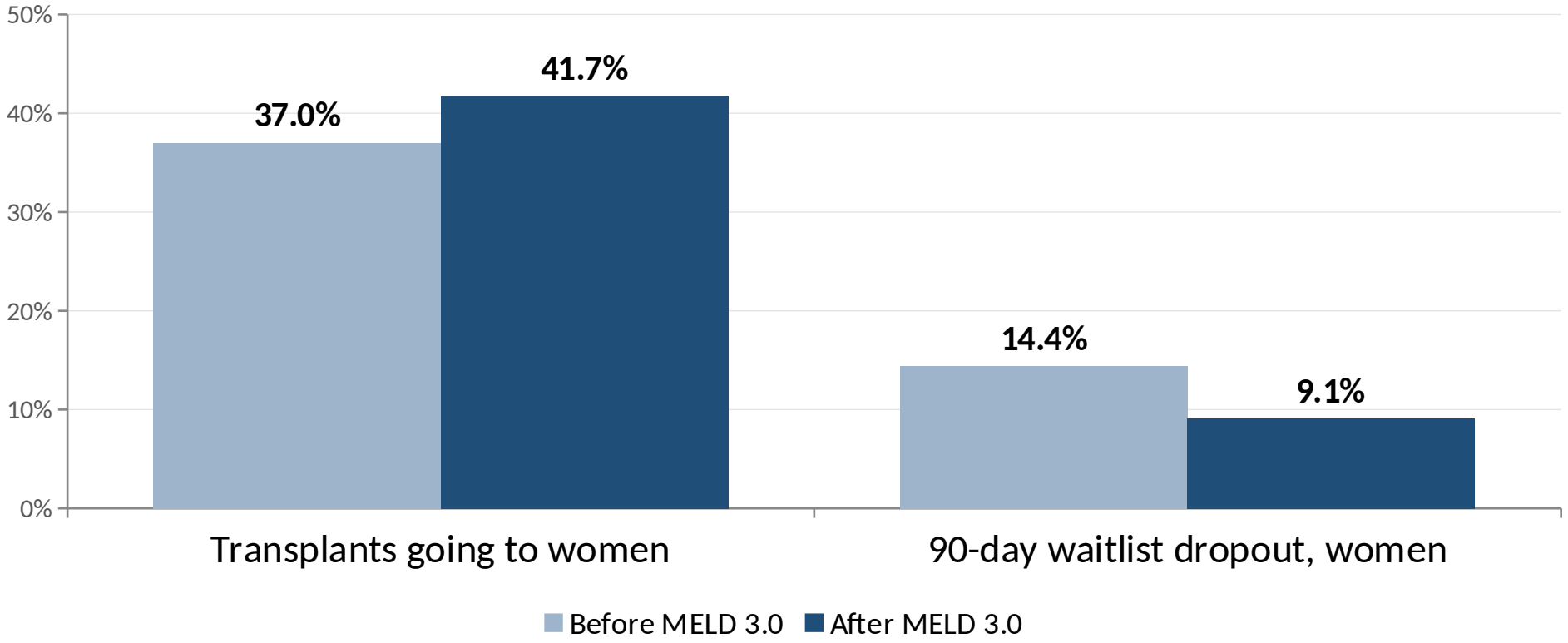
Did MELD 3.0 help women?



Conference analysis, 12 months before vs 12 months after July 2023. Dropout: died or too sick for transplant

Kwong AJ, The Liver Meeting 2024, abstract 5002 (conference data). Tanaka T et al. Hepatology 2026;84(1):106-16 [PMID: 41100798]. Di Napoli M et al. Front Transplant 2026 [PMID: 42266471]. SRTR 2024 ADR [PMID: 42276697]

Did MELD 3.0 help women?

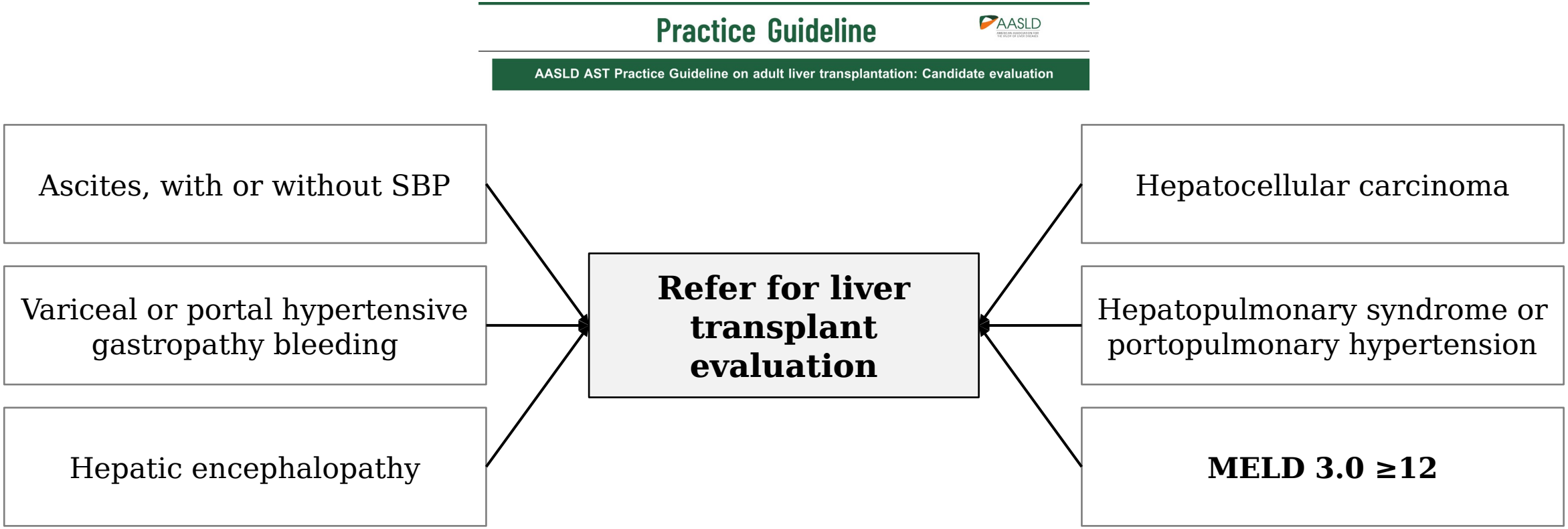


Conference analysis, 12 months before vs 12 months after July 2023. Dropout: died or too sick for transplant

The gain was largest right after the change and narrowed over time. Waitlisted women still die more often: 15.0 vs 12.4 per 100 patient-years in 2024.

Kwong AJ, The Liver Meeting 2024, abstract 5002 (conference data). Tanaka T et al. Hepatology 2026;84(1):106-16 [PMID: 41100798]. Di Napoli M et al. Front Transplant 2026 [PMID: 42266471]. SRTR 2024 ADR [PMID: 42276697]

When should a patient with cirrhosis be referred?



Statement 1: refer after any decompensating event. Without one, the MELD threshold is unclear: benefit may start at 12 in waitlisted patients, and most of the writing group still sees 15 as reasonable. “MELD score alone should not be a barrier to referral.”

Figure 1 and statement 1 redrawn from Dove L et al. AASLD/AST candidate evaluation guideline. Hepatology 2026;83(6):1609-45 [PMID: 41405234]

Who needs a call the same day?

Acute liver failure
INR ≥1.5 and any encephalopathy, no cirrhosis

Severe acute liver injury
Jaundice and a rising INR, still awake

ACLF
Cirrhosis with new organ failure

Call transplant hepatology



ALF: call “in the first few hours after presentation, and prior to meeting ALF diagnostic criteria,” because complications can make the transfer unsafe (statements 3-4)

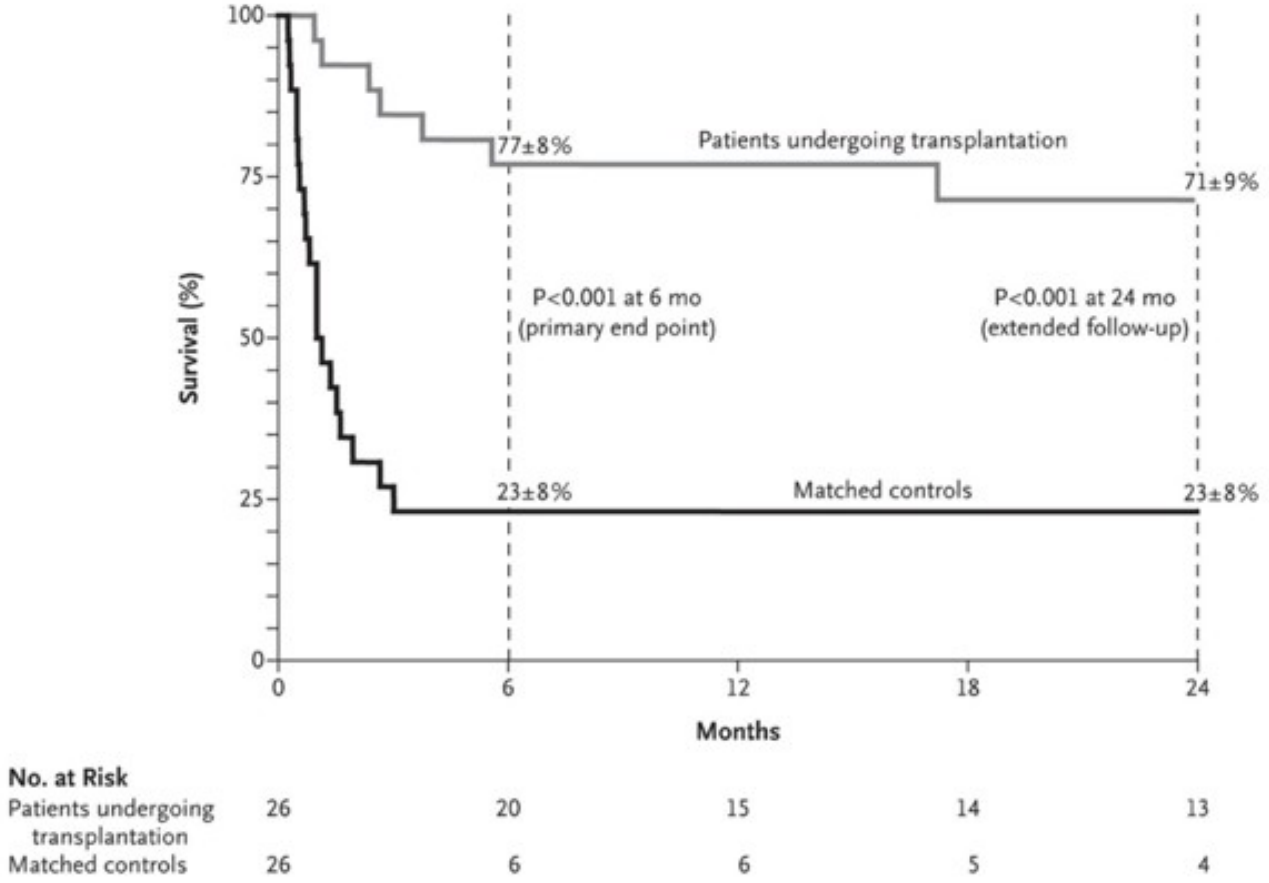
ACLF: refer (statement 2). Insurance, substance use or infection shouldn't hold up the call.

Dove L et al. AASLD/AST candidate evaluation guideline. Hepatology 2026;83(6):1609-45 [PMID: 41405234]. ACG ALF guideline 2023 [PMID: 37377263]

Who can get a liver now



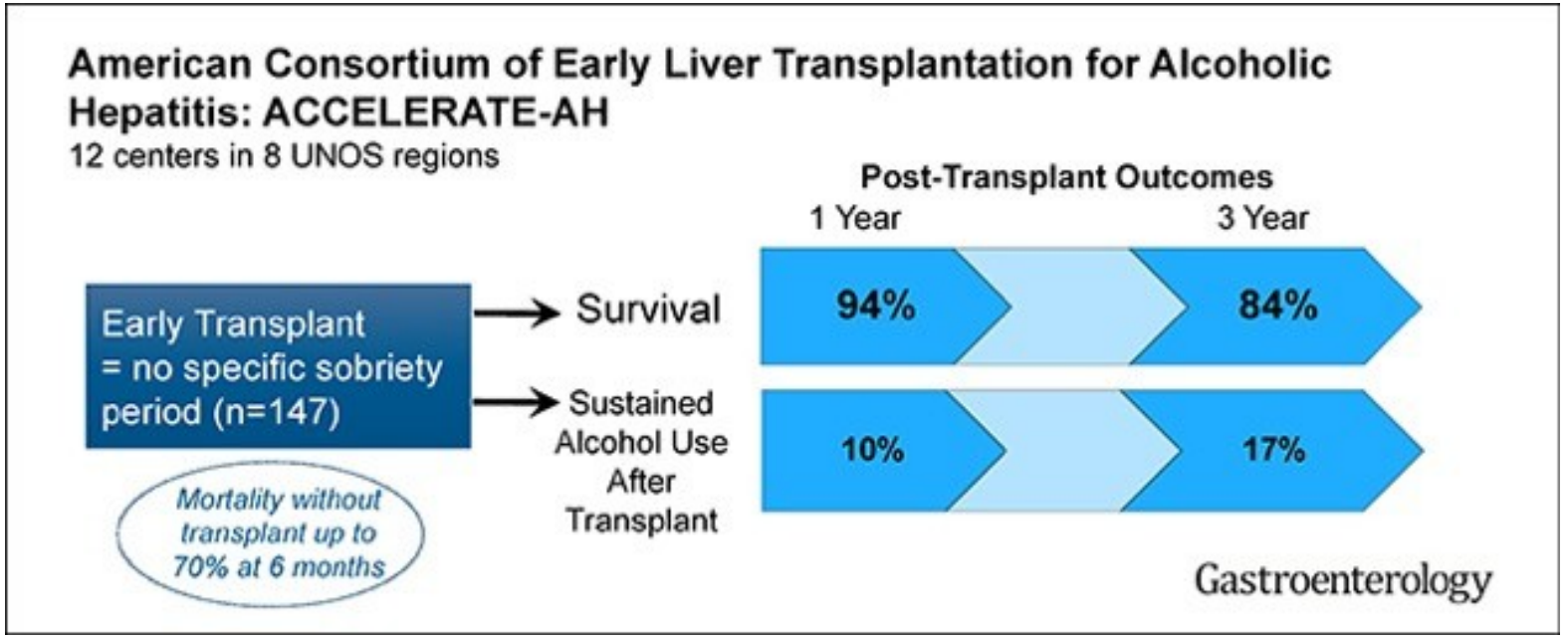
Can a patient who drank last month get a liver?



- **France and Belgium, not the US**
- First episode of severe AH, not responding to steroids, strict medical and psychosocial selection
- 26 patients, fewer than 2% of severe AH admissions
- 6-month survival 77% vs 23% in matched controls

Mathurin P, Moreno C, Samuel D, et al. N Engl J Med 2011;365(19):1790-800 [PMID: 22070476]

Did it hold up in the US?

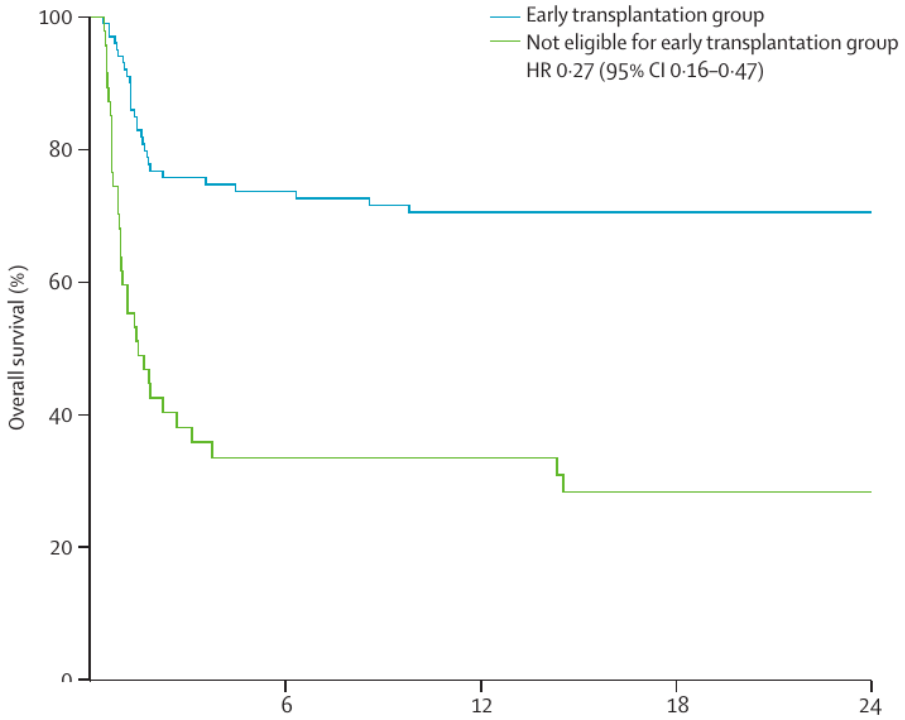


- ACCELERATE-AH, 147 patients: sustained alcohol use carried a higher risk of death (HR 4.59)
- 2025 analysis, 347 patients alive at discharge: 5-year survival 83.1%

Lee BP et al. Gastroenterology 2018;155(2):422-430.e1 [PMID: 29655837]. Dukewich M et al. Am J Gastroenterol 2025;120(4):827-36 [PMID: 38994850]

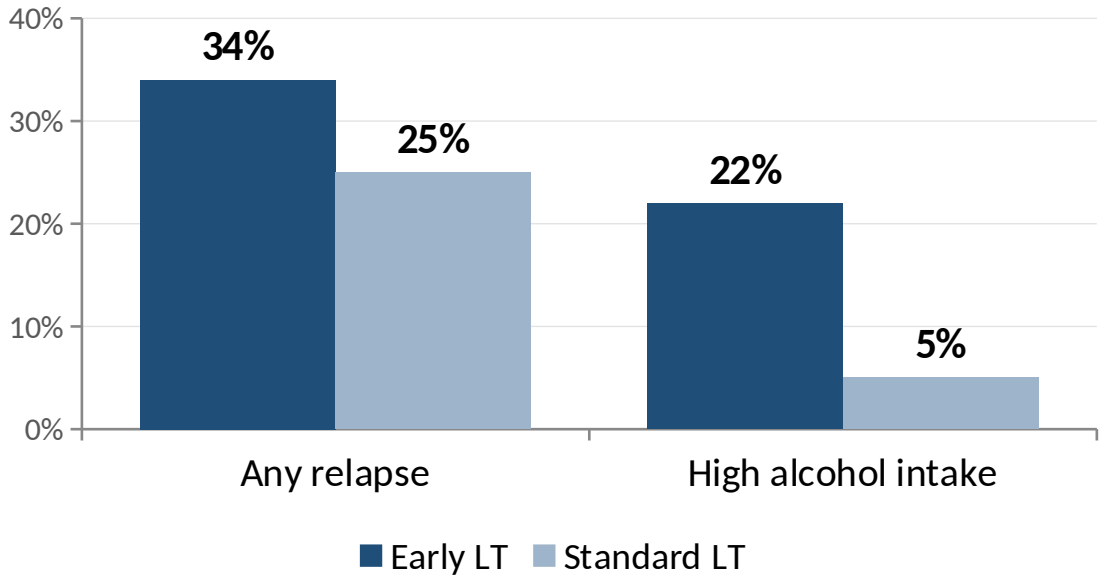
What about relapse?

Early liver transplantation for severe alcohol-related hepatitis not responding to medical treatment: a prospective controlled study



Early-LT strategy vs patients not eligible: 2-year survival 70.6% vs 28.3%

Relapse in transplanted patients: early vs standard LT



Non-inferiority for relapse not shown

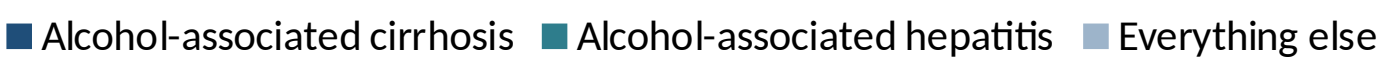
Louvet A, Labreuche J, Moreno C, et al. Lancet Gastroenterol Hepatol 2022;7(5):416-25 [PMID: 35202597]

What do the guidelines say now?

Guideline	Early transplant for alcohol-associated hepatitis
EASL 2024	Selected non-responders with favorable sobriety predictors (strong)
ACG 2024	Carefully selected non-responders at low risk of relapse
AASLD/AST 2025	Refer ALD early; abstinence duration alone doesn't exclude (37-38)

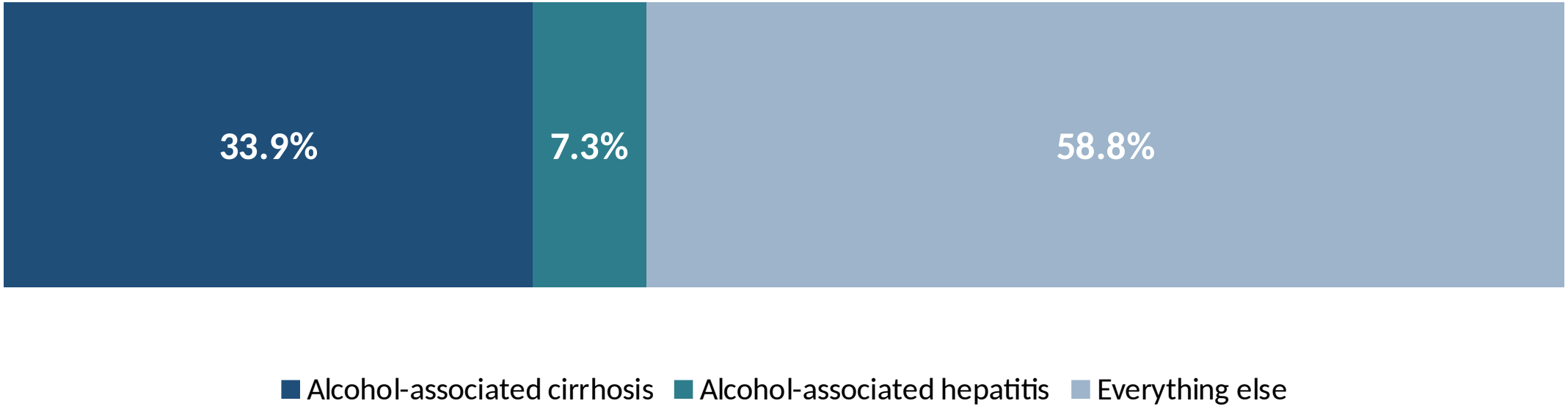
EASL CPG on liver transplantation. J Hepatol 2024;81(6):1040-86, p.1044 [PMID: 39487043]. Jophlin LL et al. Am J Gastroenterol 2024;119(1):30-54 [PMID: 38174913]. Dove L et al. Hepatology 2026;83(6):1609-45 [PMID: 41405234]

What do US adults get a liver for?



OPTN/SRTR 2024 Annual Data Report, Table LI 7. Kwong AJ et al. OPTN/SRTR 2024 Annual Data Report: Liver. Am J Transplant 2026;26(8S1):S214-S320 [PMID: 42276697]

What do US adults get a liver for?

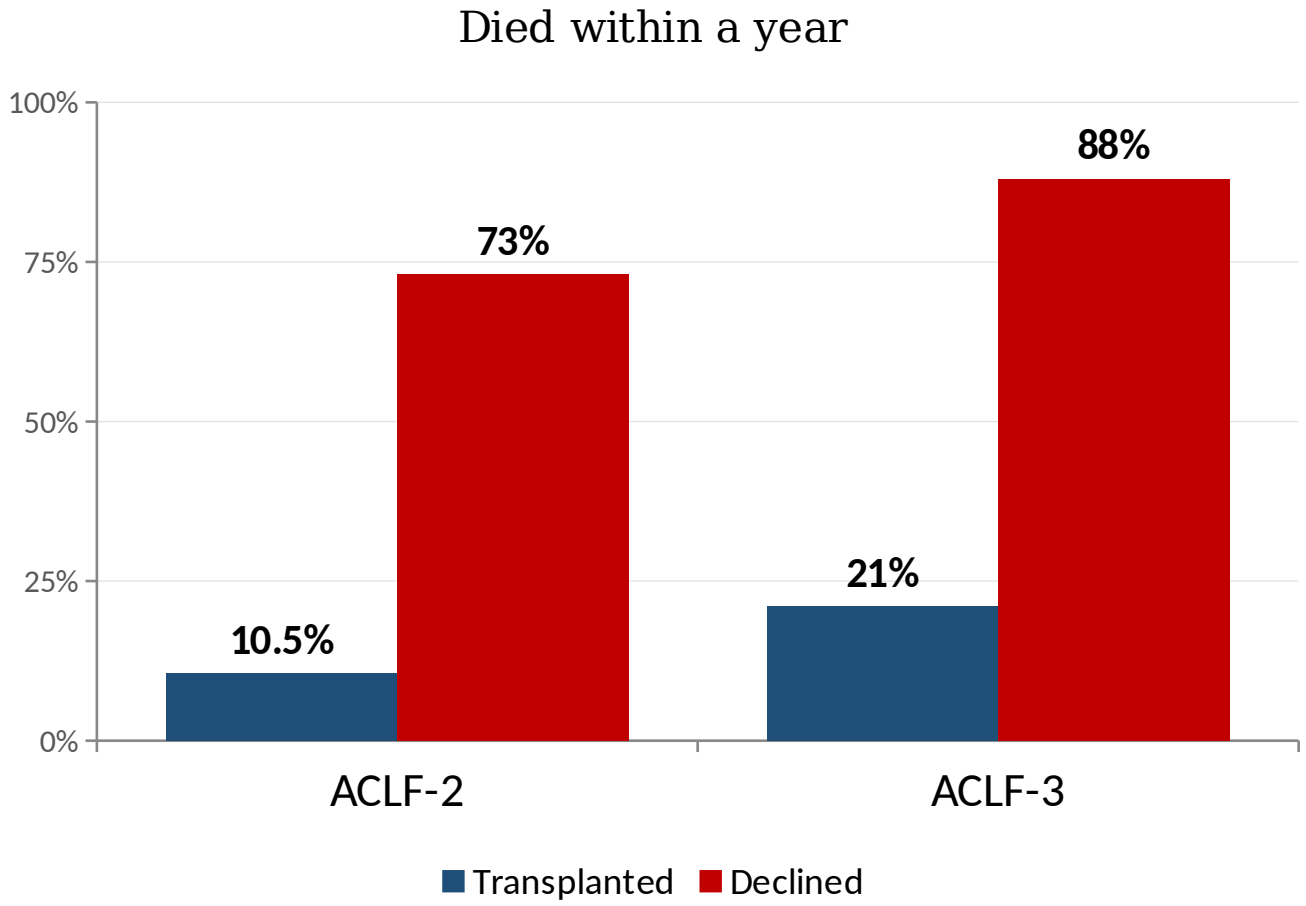


Alcohol-associated liver disease: 41.1% of US adult recipients, the top diagnosis

Parts are rounded, so they add to 41.2%

OPTN/SRTR 2024 Annual Data Report, Table LI 7. Kwong AJ et al. OPTN/SRTR 2024 Annual Data Report: Liver. Am J Transplant 2026;26(8S1):S214-S320 [PMID: 42276697]

Can patients with ACLF get a liver?



Transplanted: case-fatality, no retransplant. Declined: 1-year mortality

- CHANCE: 612 transplanted, 144 declined, 63 centers
- Without ACLF: 9% case-fatality at a year after transplant
- AASLD 2024: selected patients with ACLF may warrant expedited transplant; palliative care can run alongside the evaluation

Gustot T, Bernal W, Fernández J, et al. CHANCE. J Hepatol 2026 [PMID: 42822699]. Karvellas CJ et al. Hepatology 2024;79(6):1463-1502, statements 47, 51 [PMID: 37939273]

What no longer rules a patient out?

On its own	2025 AASLD/AST guideline	Statement
Length of abstinence	Duration alone doesn't exclude	38
Methadone or buprenorphine	No denial, no taper needed	41
BMI >40 or <18.5	Screen nutrition and muscle; BMI alone doesn't exclude	48-49
ACLF or MELD >40	Doesn't exclude by itself	2, 43
Older age (living donor LT)	Not an exclusion	51

Each one is still weighed with the rest of the picture, case by case.

Dove L et al. AASLD/AST candidate evaluation guideline. Hepatology 2026;83(6):1609-45 [PMID: 41405234]

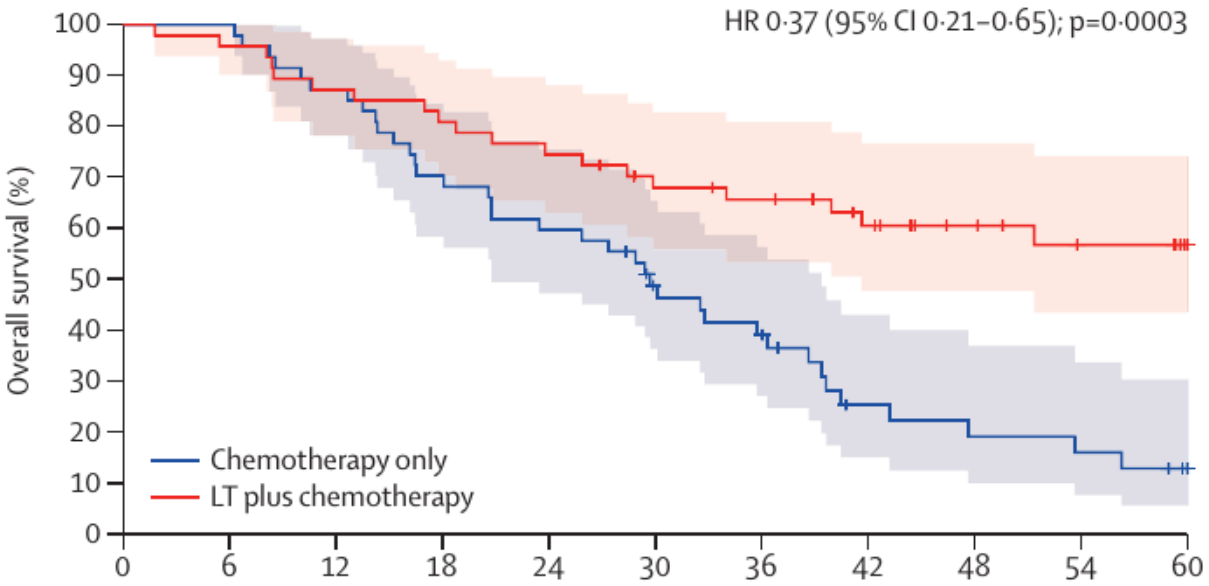
Who gets exception points?

Exception	Criteria	Score
HCC	1 lesion 2-5 cm, or 2-3 lesions each 1-3 cm AFP ≤1000 (if >1000, must fall below 500)	MELD, then MMaT-3
Hepatopulmonary syndrome	Confirmed HPS, room-air PaO2 <60 mmHg	MMaT-3
Portopulmonary HTN	Before treatment: mPAP ≥35, PVR ≥3 WU After: mPAP <35, PVR <5; or 35 to <45, PVR <3	MMaT-3
Colorectal liver metastases	Non-standard exception, reviewed	MMaT-20 (floor 15)
Intrahepatic cholangiocarcinoma	Non-standard exception, reviewed	MMaT-3

OPTN Policies 9.5.E, 9.5.G, 9.5.I (effective 10/1/2026); OPTN adult MELD exception transplant oncology review guidance

Can a liver transplant help colorectal metastases?

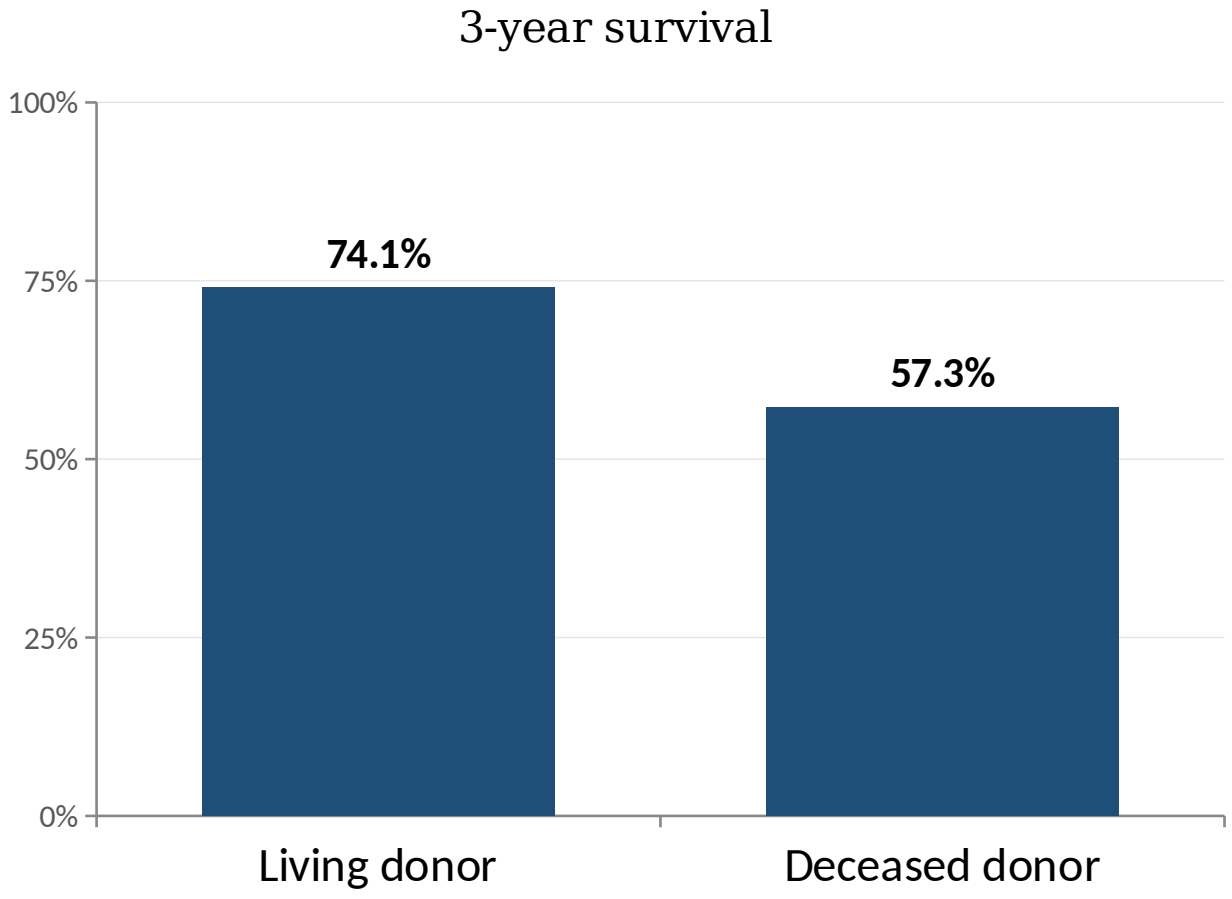
Liver transplantation plus chemotherapy versus chemotherapy alone in patients with permanently unresectable colorectal liver metastases (TransMet): results from a multicentre, open-label, prospective, randomised controlled trial



- **20 centers in France, Belgium and Italy, not the US**
- 94 randomized: age 18-65, BRAF wild-type, primary resected, permanently unresectable liver-only disease responding to chemotherapy
- 5-year survival (intention to treat): 56.6% with LT plus chemotherapy vs 12.6% with chemotherapy
- 72% of transplanted patients recurred, mostly in the lungs; almost half of recurrences were resected or ablated

Adam R, Piedvache C, Chiche L, et al. Lancet 2024;404(10458):1107-18 [PMID: 39306468]

And in the US?



Registry comparison, not adjusted

- 158 transplants for colorectal metastases at 33 US centers, Dec 2017 to Mar 2025
- About half from living donors
- 3-year survival 66.7%, disease-free 44.0%
- Exception points by review since Feb 2025

Chávez-Villa M, Sasaki K, Nakayama T, et al. Liver Transpl 2026;32(4):508-17 [PMID: 41086433]

From Maine to Boston



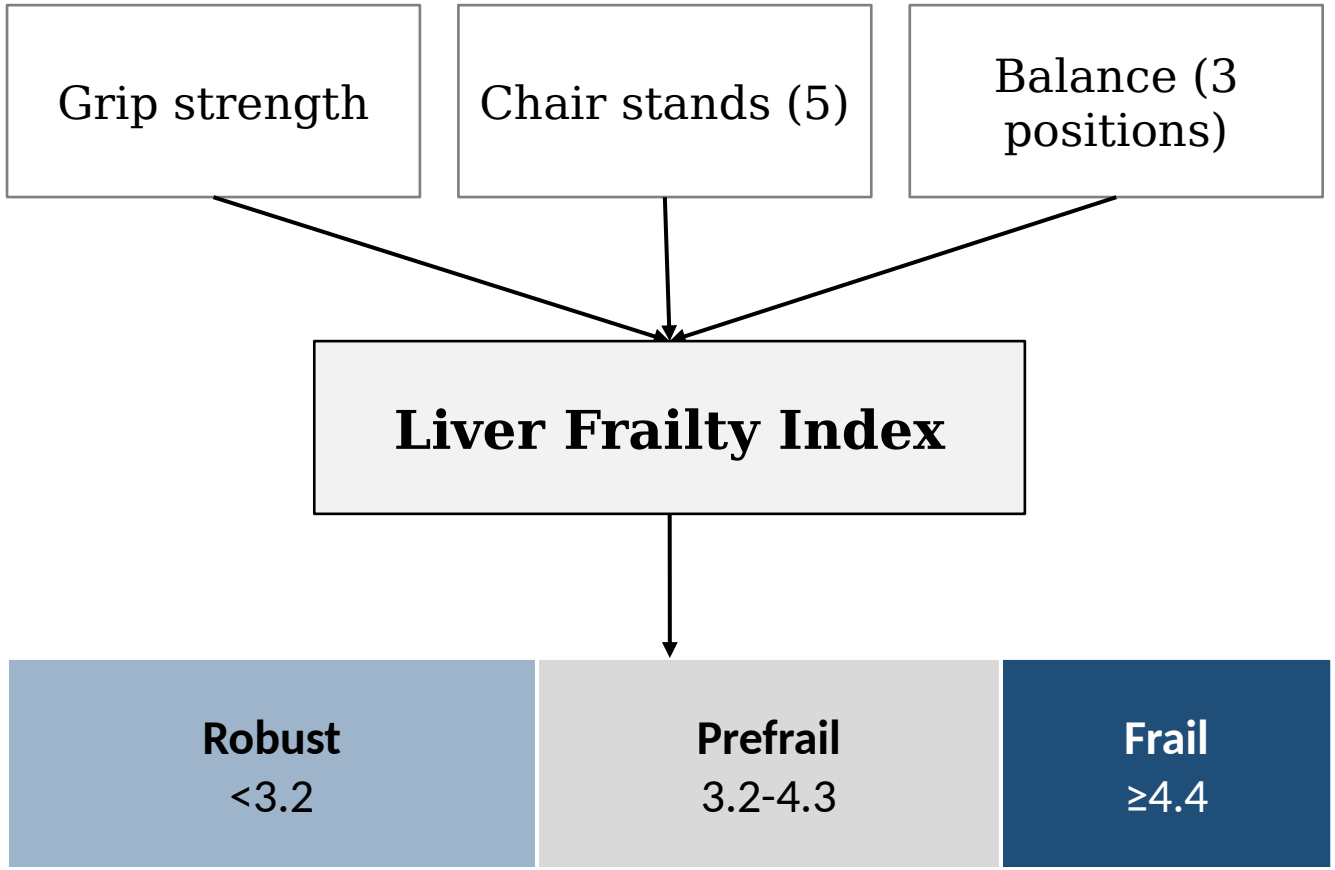
What happens after the referral?



Part of the evaluation	What it involves
Hepatology and surgery	Visits, labs, CT or MRI of the liver
Heart and lungs	ECG and echo, then coronary testing by risk
Social work and addiction	Home support, alcohol and drug use, treatment plan
Nutrition and physical therapy	Muscle, frailty, protein intake
Other	Dental, cancer screening, vaccines, financial counseling

Dove L et al. AASLD/AST candidate evaluation guideline. Hepatology 2026;83(6):1609-45 [PMID: 41405234]

How should frailty change the evaluation?

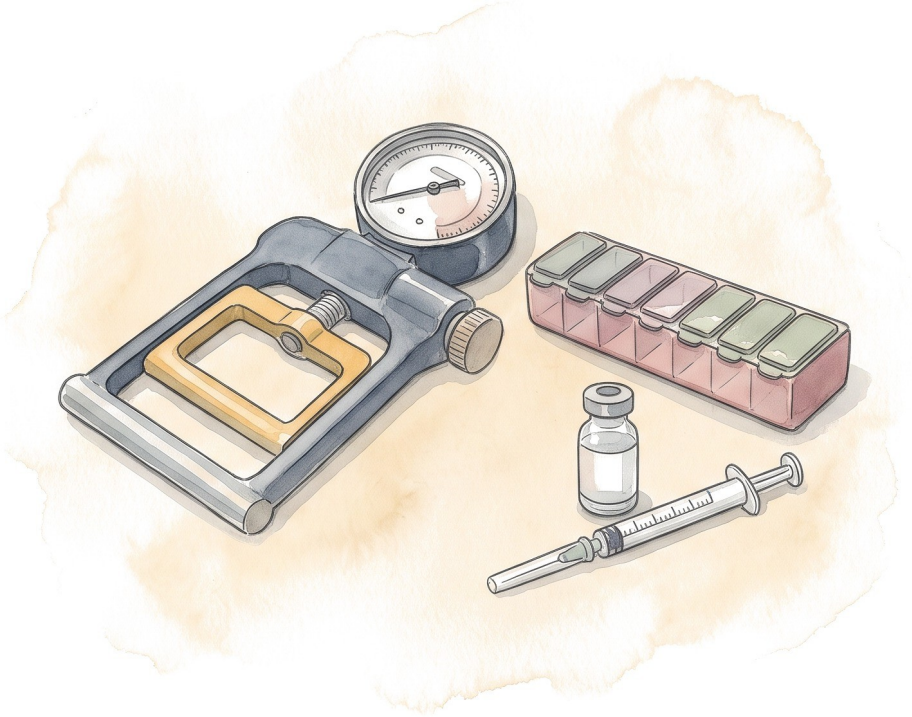


- Measure it with a standard tool (statement 33)
- Frailty raises the risk after transplant, but alone it doesn't rule a patient out (44-45)
- Frailty also predicts death on the waitlist

Lai JC, Tandon P, Bernal W, et al. Hepatology 2021;74(3):1611-44 [PMID: 34233031]. Dove L et al. Hepatology 2026;83(6):1609-45 [PMID: 41405234]

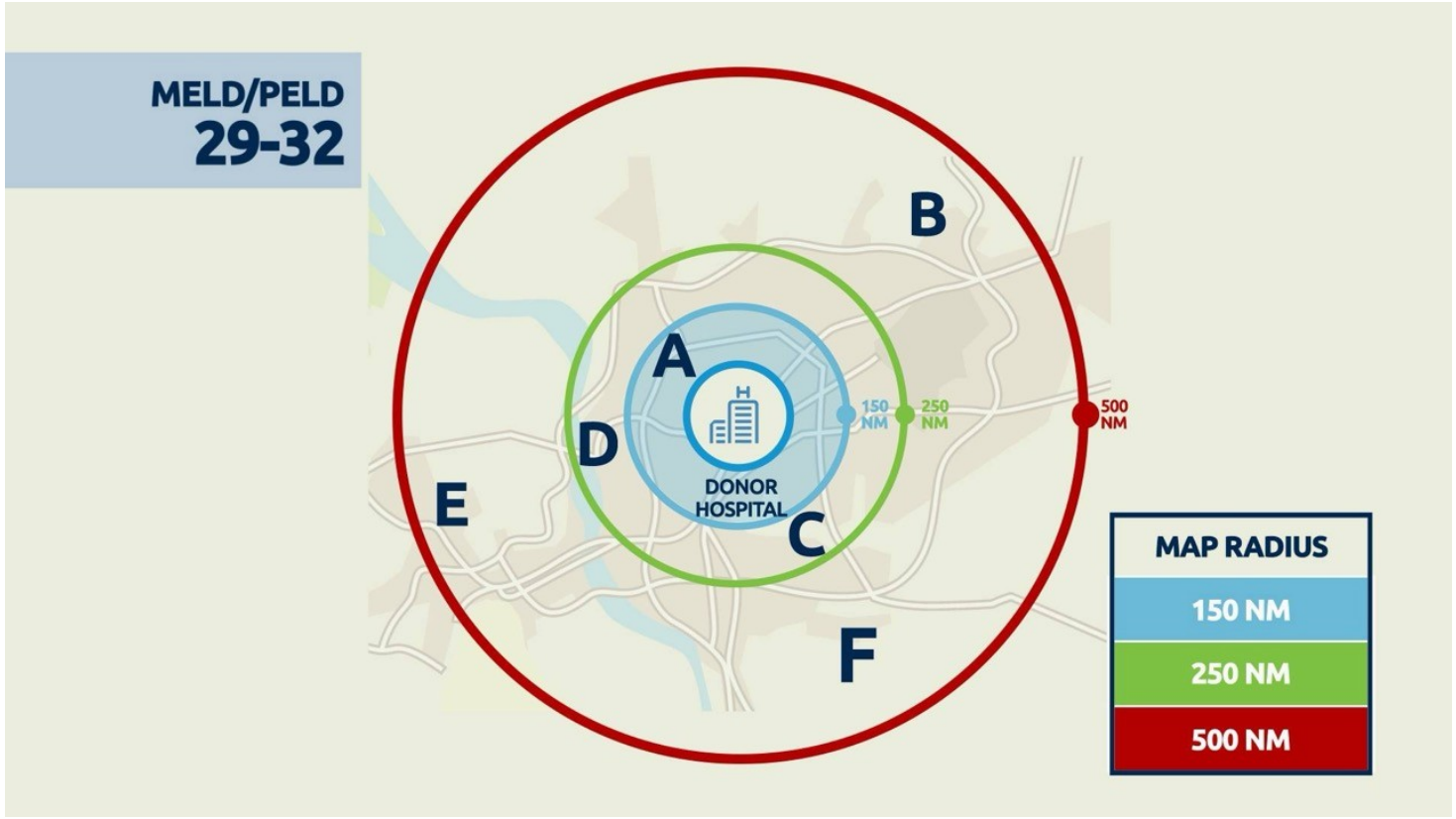
What can you do while the patient waits?

- Treat alcohol use disorder, with medicines and counseling
- Protein 1.2-1.5 g/kg ideal body weight a day, a late-evening snack
- Exercise for frailty
- Vaccines before transplant
- No known HCC: surveillance every 6 months
- Keep the MELD labs current



Dove L et al. Hepatology 2026;83(6):1609-45 [PMID: 41405234]. Lai JC et al. Hepatology 2021 [PMID: 34233031]

How does a liver get from a donor to a patient?



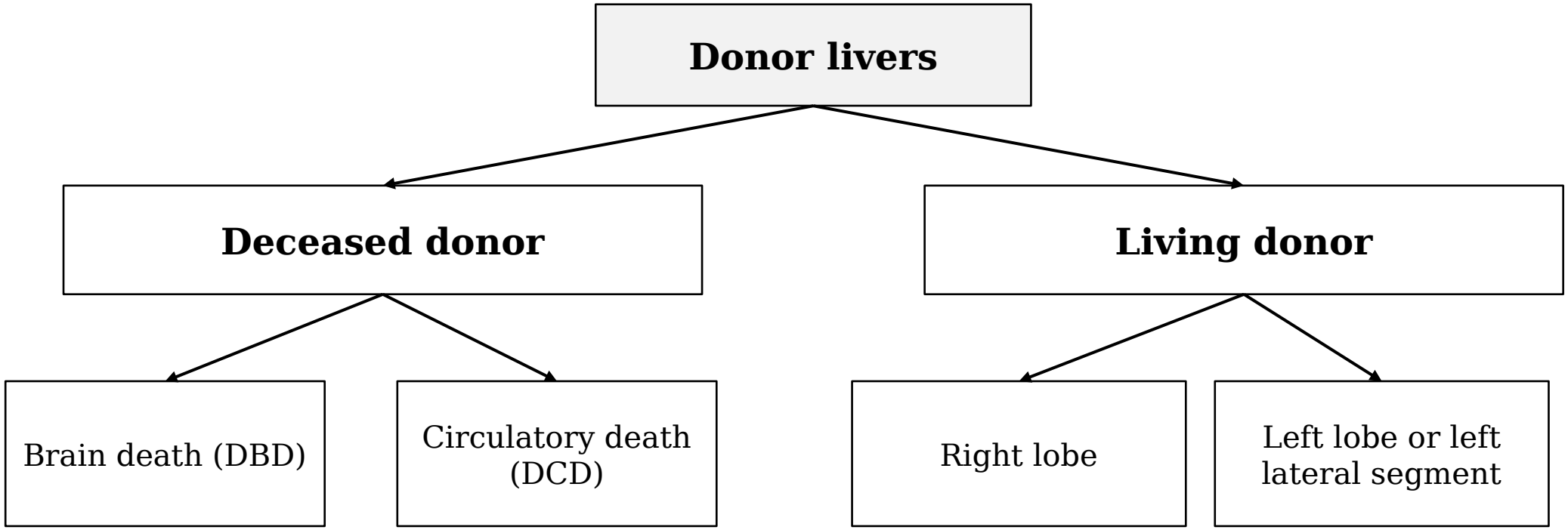
- Circles of 150, 250 and 500 nautical miles around the donor hospital
- Offers follow urgency first, then distance from the donor hospital
- Exception points are set from the median MELD at transplant (MMaT) near the donor hospital

OPTN Policy 9, liver allocation (acuity circles since February 2020)

Where the livers come from



What kinds of donors are there?



Either can go whole or be split

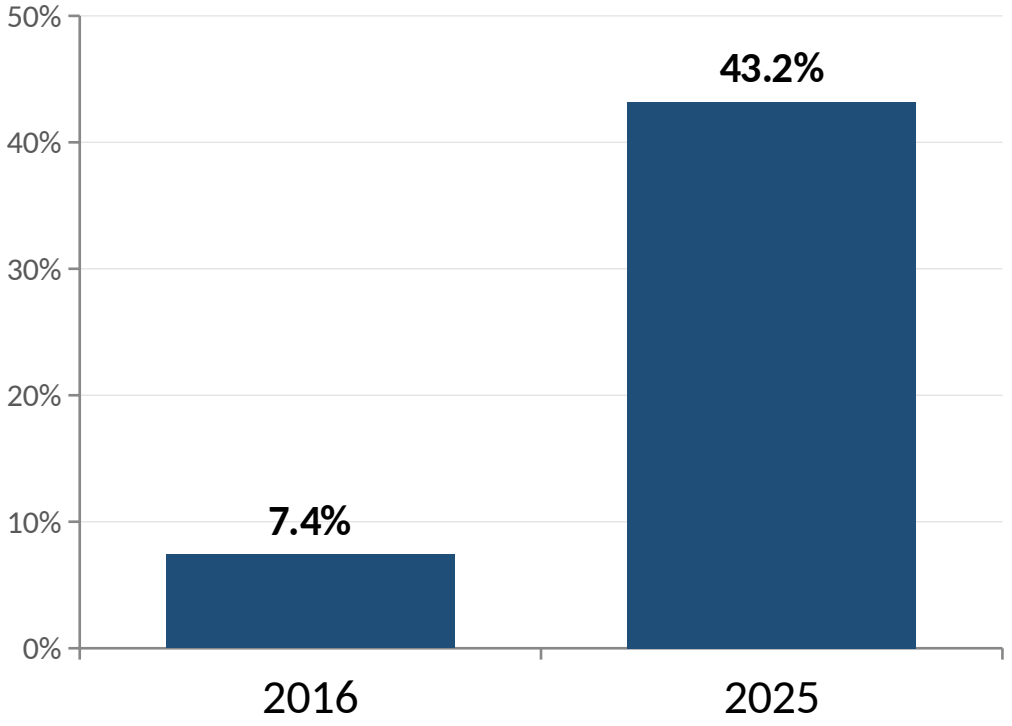
DCD: 43% of US deceased donor liver transplants in 2025

Living donor: 4.5% of US adult liver transplants in 2024

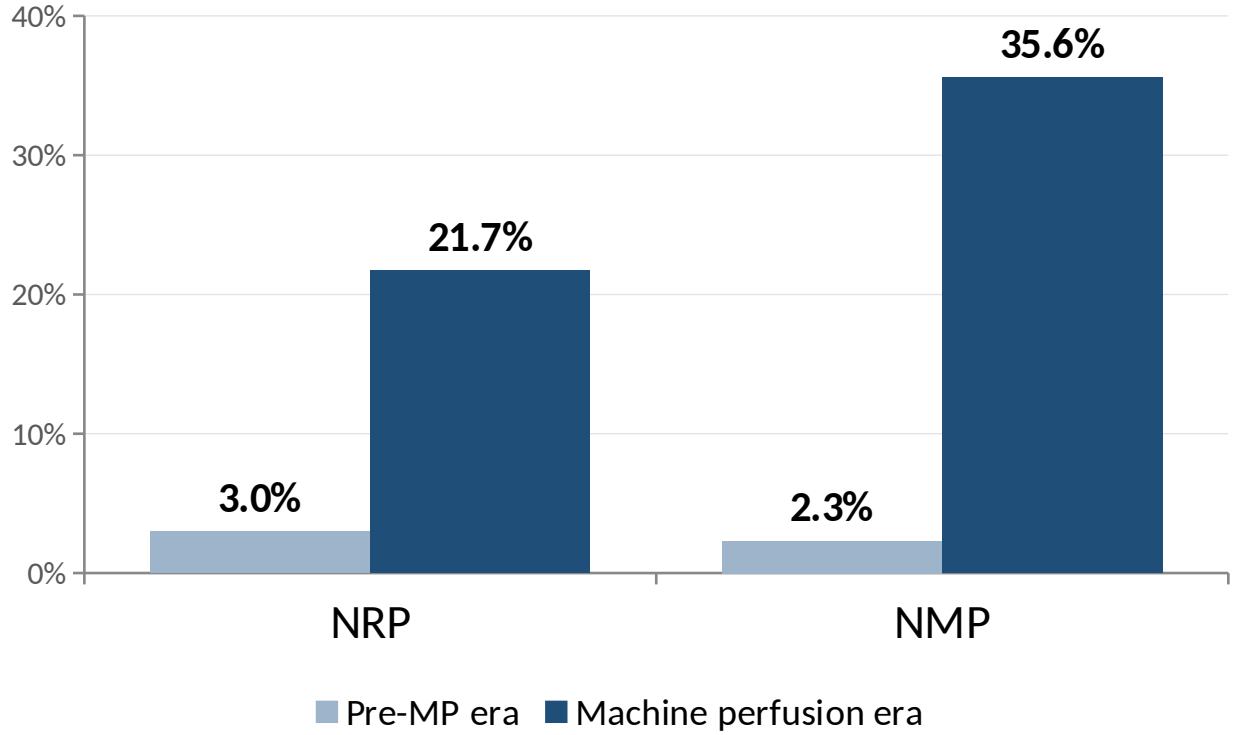
Giorgakis E et al. Transpl Int 2026;39:16689 [PMID: 42294094]. Kwong AJ et al. OPTN/SRTR 2024 Annual Data Report: Liver. Am J Transplant 2026;26(8S1):S214-S320 [PMID: 42276697]

How much has DCD grown?

DCD share of deceased donor LTs



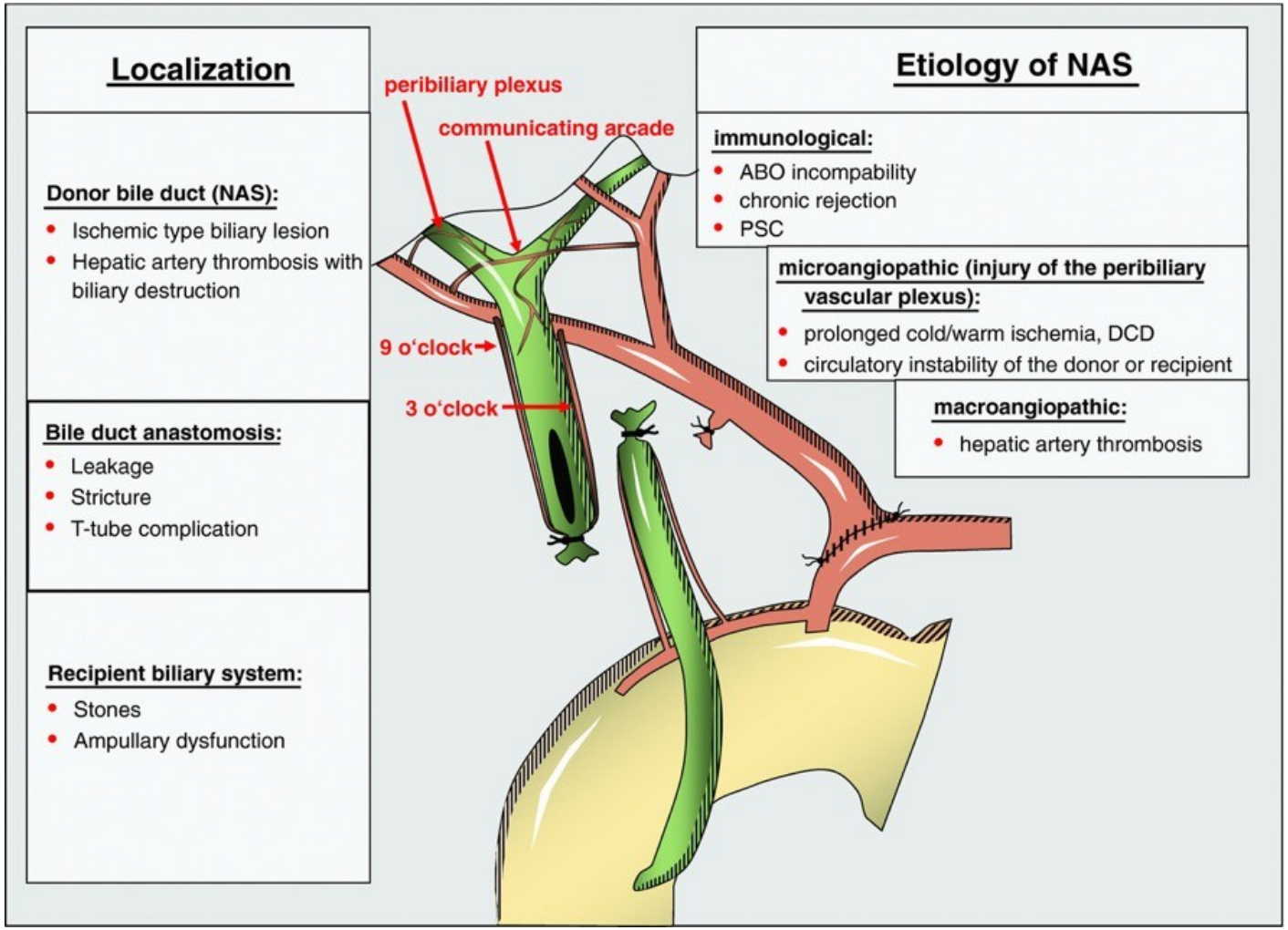
Use of NRP and NMP



Adult DCD recipients: median wait 112 vs 62 days, earlier vs later era

Giorgakis E, Ziogas IA, Moris D, et al. Transpl Int 2026;39:16689 [PMID: 42294094]

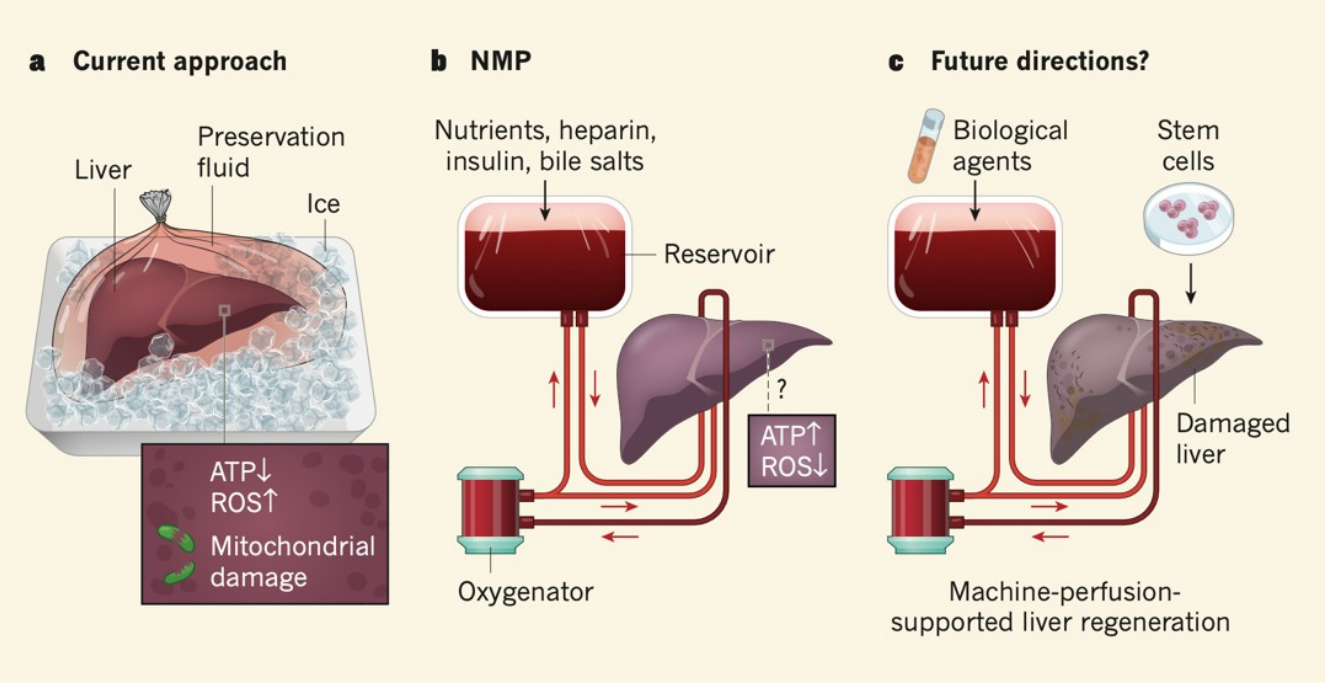
Why did DCD livers worry us?

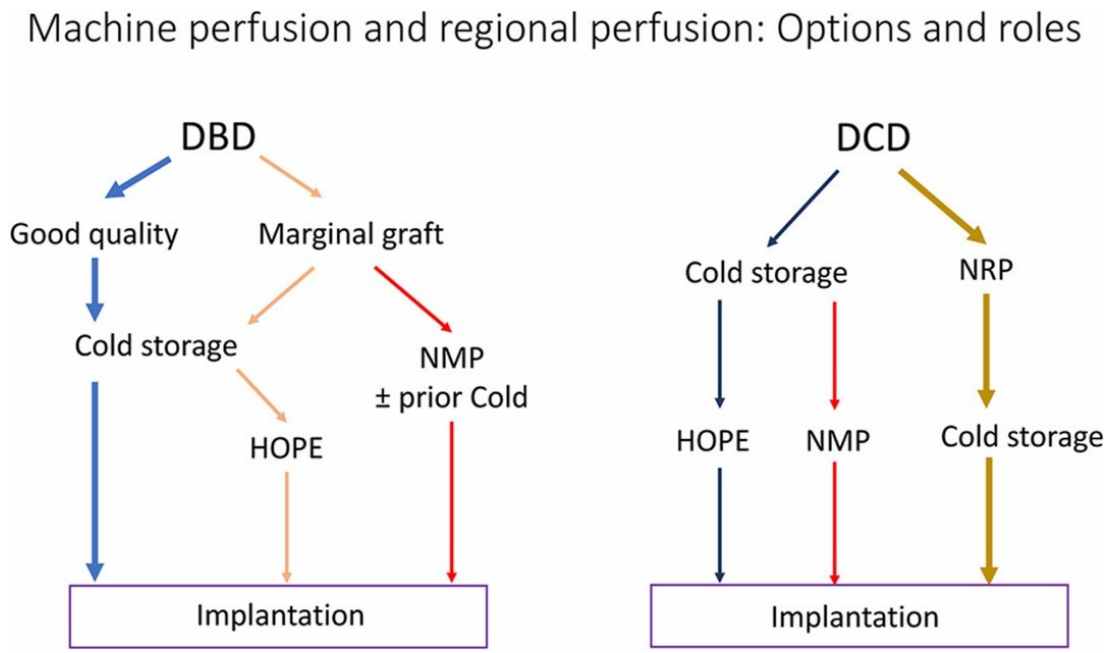


- After transplant, the donor bile ducts depend on the hepatic artery, through the peribiliary plexus
- Warm ischemia during DCD injures that plexus and the duct lining
- Weeks to months later: non-anastomotic strictures, sometimes needing retransplant

Seehofer D, Eurich D, Veltzke-Schlieker W, Neuhaus P. Am J Transplant 2013;13(2):253-65 [PMID: 23331505]

What does a perfusion machine do?



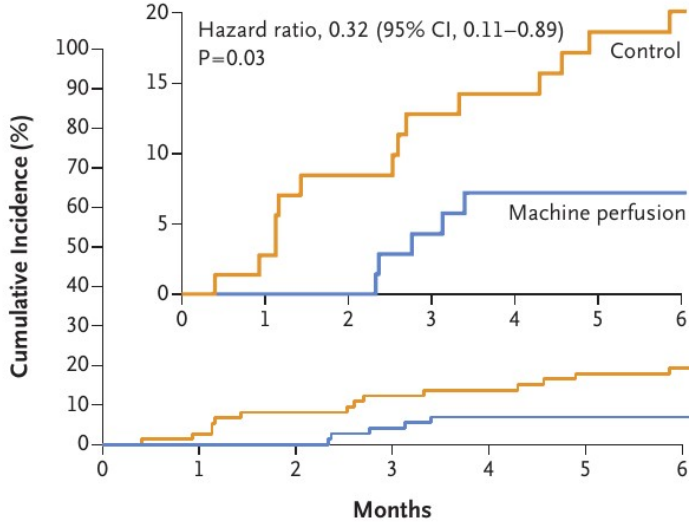


- **HOPE:** cold and oxygenated, usually at the transplant center just before implant
- **NMP:** warm and blood-based; the liver makes bile and can be assessed before use
- **NRP:** oxygenated blood flow restored to the donor's abdominal organs after death is declared

Schneeberger S. Nature 2018;557(7703):40-1 [PMID: 29713063]. J Clin Exp Hepatol 2024;14(2):101309 [PMID: 38274508]

What did the trials show?

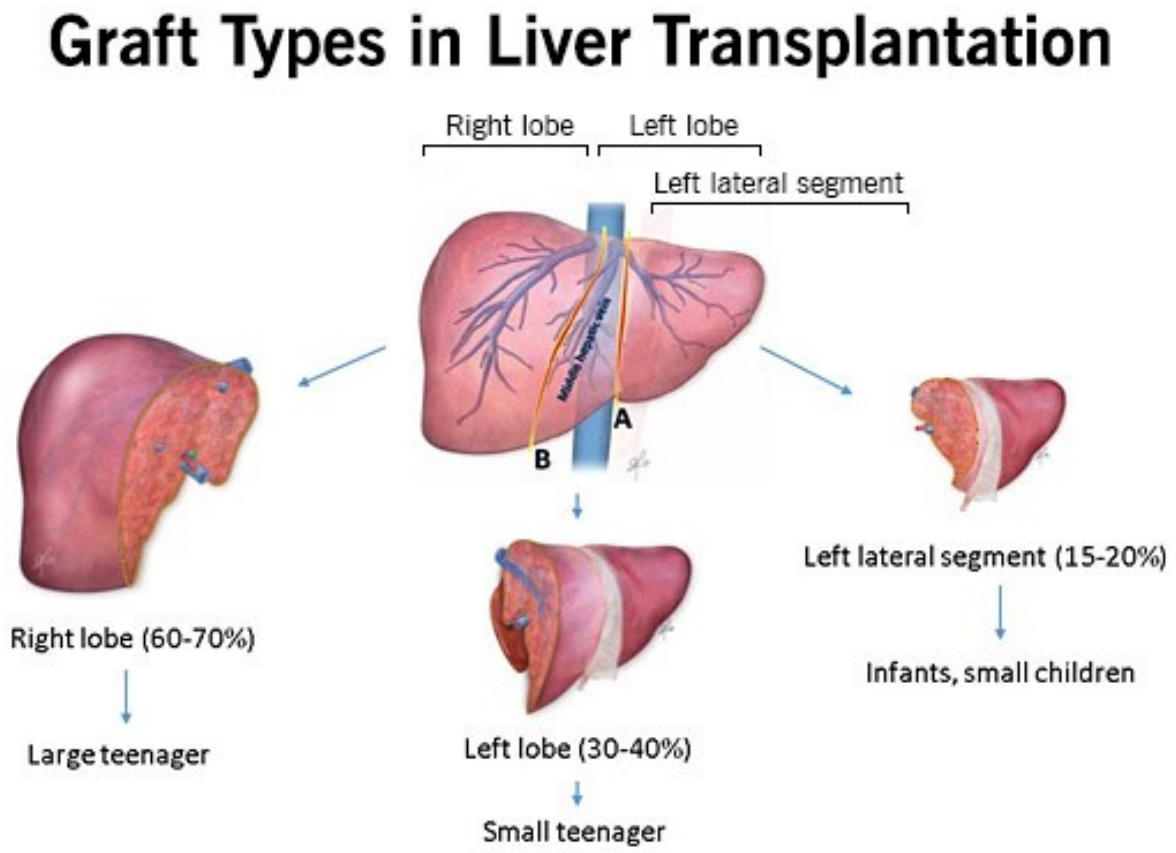
Trial	Machine	Main result
Nasralla, Nature 2018 (220 LTs)	NMP	Peak AST and discard 50% lower
PROTECT, JAMA Surg 2022	NMP (OCS)	EAD 18% vs 31% Ischemic biliary 2.6% vs 9.9%
DHOPE-DCD, NEJM 2021	DHOPE	Symptomatic NAS 6% vs 18% at 6 months (RR 0.36)



The 2025 AASLD/AST and 2024 EASL guidelines both recommend machine perfusion for DCD and extended-criteria livers.

Nasralla D et al. Nature 2018 [PMID: 29670285]. Markmann JF et al. JAMA Surg 2022 [PMID: 34985503]. van Rijn R et al. N Engl J Med 2021 [PMID: 33626248]. Te HS et al. 2026, statements 6-10 [PMID: 40844852]. EASL 2024, p.1062

What about living donors?



- 4.5% of US adult liver transplants in 2024
- Aim for a graft-to-recipient weight ratio of 0.8% or more
- Caution above MELD 30
- Older recipient age alone doesn't exclude

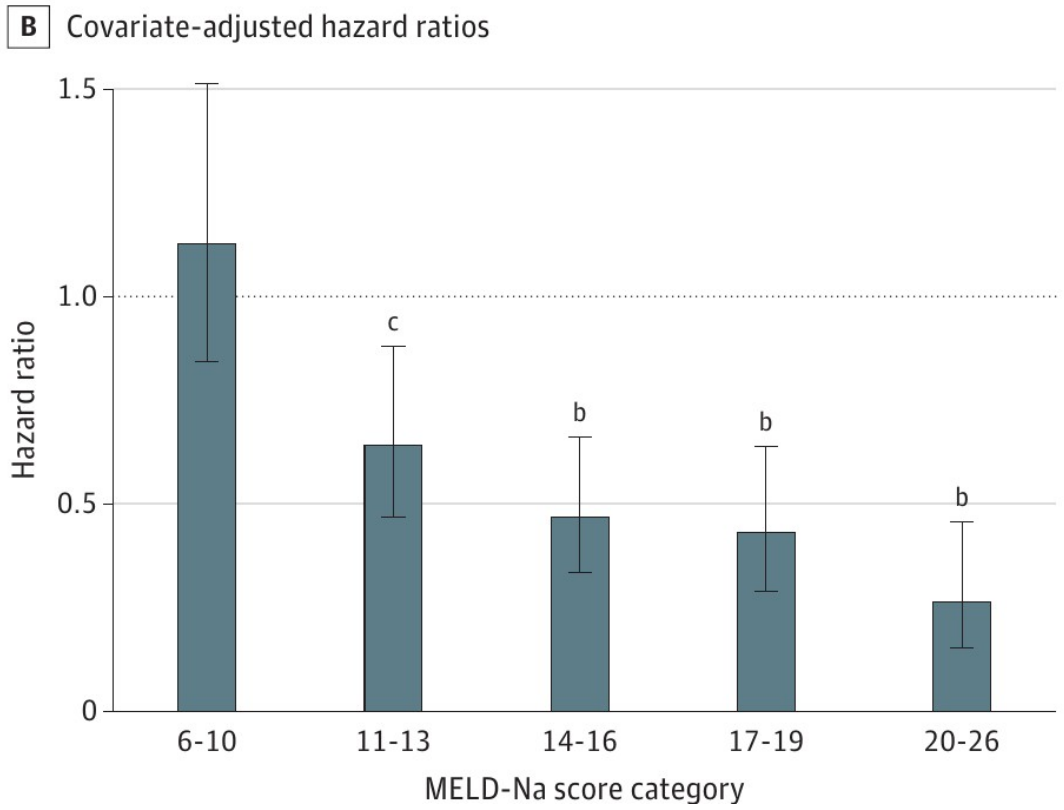
Dove L et al. Hepatology 2026;83(6):1609-45 [PMID: 41405234], statements 50-53. Kwong AJ et al. OPTN/SRTR 2024 Annual Data Report: Liver. Am J Transplant 2026;26(8S1):S214-S320 [PMID: 42276697]

Is there benefit below MELD 15 with a living donor?

JAMA Surg. 2022 Oct 1;157(10):926-932. doi: 10.1001/jamasurg.2022.3327.

Survival Benefit of Living–Donor Liver Transplant

Whitney E Jackson [1 2], John S Malamon [2 3], Bruce Kaplan [1 2 3], Jessica L Saben [2 3], Jesse D Schold [4], James J Pomposelli [2 3], Elizabeth A Pomfret [2 3]

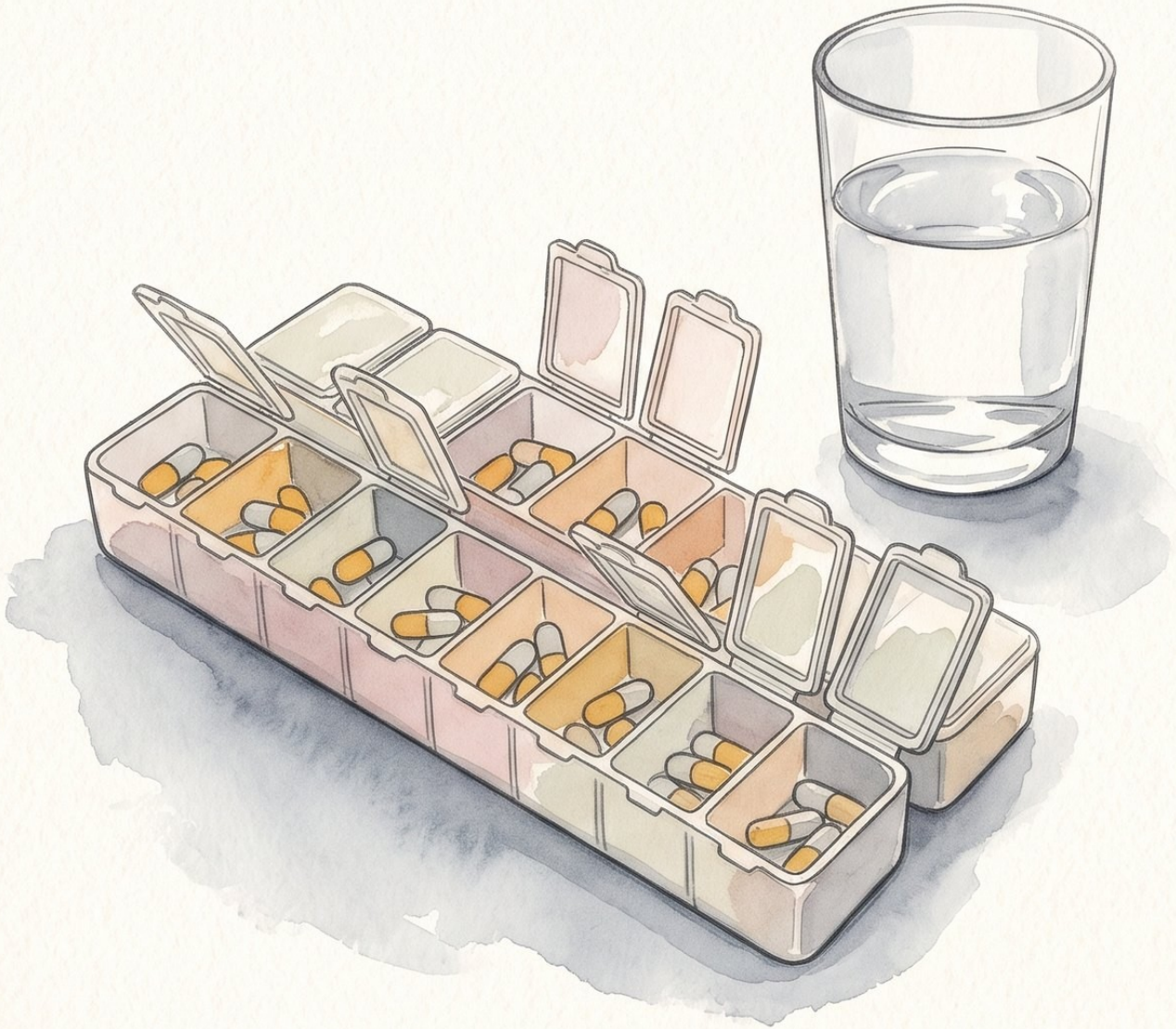


Adjusted, compared with staying on the list:

- MELD-Na 11-13: aHR 0.64 (0.47-0.88)
- MELD-Na 14-16: aHR 0.47 (0.34-0.66)
- Modeled gain of 13-17 life-years

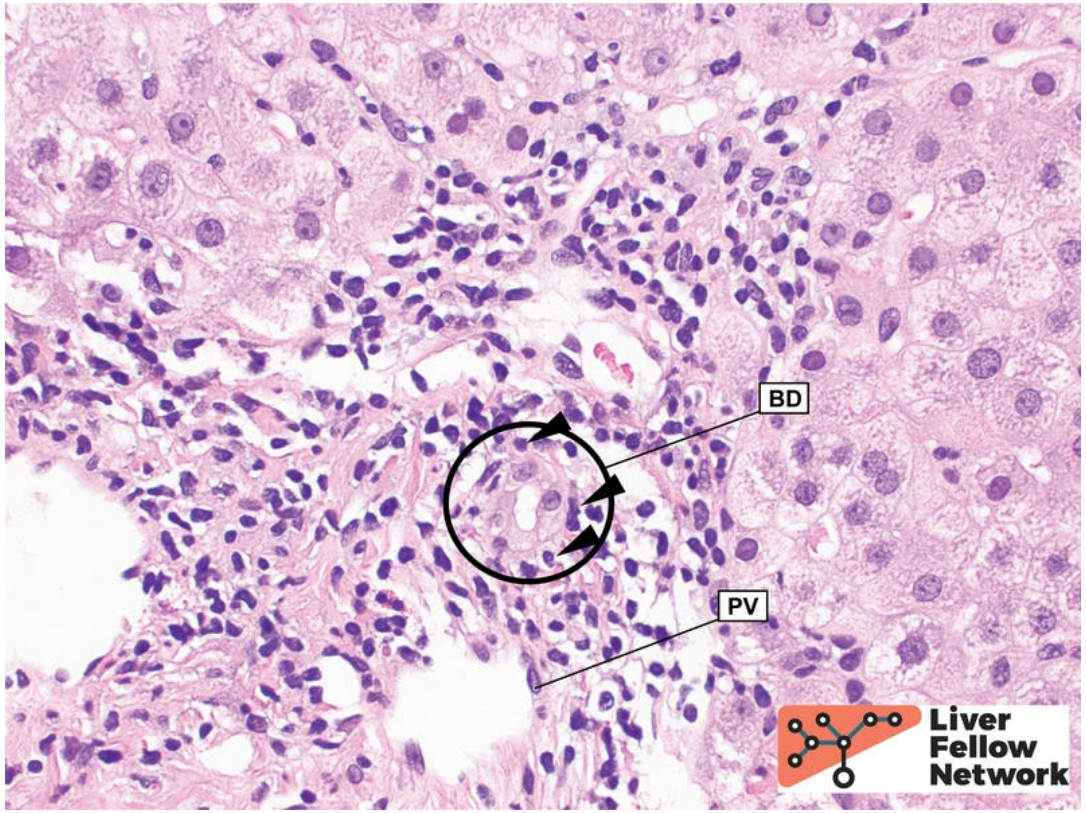
Jackson WE, Malamon JS, Kaplan B, Saben JL, Schold JD, Pomposelli JJ, Pomfret EA. JAMA Surg 2022;157(10):926-932 [PMID: 35921119]

Your patient had a transplant



Abnormal liver tests after transplant: what next?

- New or rising liver tests: call transplant hepatology the same day
- Look at the timing, the pattern, new drugs and old labs
- Doppler ultrasound and a correctly timed tacrolimus trough
- Think vascular, biliary, infection, drugs and rejection
- The transplant team decides on a biopsy



Acute cellular rejection

MRCP or ERCP for a suspected stricture?



Situation	Next step
Suspected stricture	**MRCP first (strong)**
High suspicion, or intervention likely	ERCP directly (weak)
Normal MRCP, picture still fits	ERCP (weak)
Progressive ischemic cholangiopathy	Consider retransplant

Te HS et al. AASLD/AST graft-related complications guideline. Liver Transpl 2026;32(3):444-90 [PMID: 40844852], statements 15, 16, 24. Kohli DR et al. ASGE guideline. Gastrointest Endosc 2023;97(4):607-14 [PMID: 36797162]

Which drugs change tacrolimus levels?

Raise levels (toxicity)	Lower levels (rejection)
Fluconazole, voriconazole, posaconazole	Rifampin
Clarithromycin, erythromycin	Phenytoin, carbamazepine
Diltiazem, verapamil	Phenobarbital
Nirmatrelvir/ritonavir (Paxlovid)	St John's wort
Ritonavir- or cobicistat-boosted HIV drugs	
Grapefruit, cannabidiol (CBD)	

Starting or stopping a drug, or the patient can't take pills: call transplant pharmacy or hepatology for a plan.

Tacrolimus (Prograf) prescribing information. Sharma P et al. AASLD/AST non-graft complications guideline. Liver Transpl 2026;32(8):1157-93 [PMID: 41405235]

What does the primary care doctor need to do?

	2025 guideline	Statement
Screening	BP, weight, glucose and lipids every year	53
Type 2 diabetes	Metformin; GLP-1 RA or SGLT2i first if high CV, HF or CKD risk	58-59
Hypertension	Amlodipine or felodipine; ACEi or ARB if CVD or CKD	60-61
Lipids	Hydrophilic statin first	62
Obesity	GLP-1 RA if diet and exercise aren't enough	64
CKD	Consider nephrology at stage 4	76, 79, 82

Sharma P et al. AASLD/AST non-graft complications guideline. Liver Transpl 2026;32(8):1157-93 [PMID: 41405235]

What else does the primary care doctor need to do?

	2025 guideline	Statement
Cancer	Age-appropriate screening; skin exam every year	44-45
Vaccines	ACIP schedule for immunocompromised adults; no live vaccines	19-20
Alcohol	None, whatever the original liver disease	9
Elective surgery	At least 3 months after LT, with stable graft function	83-84
Pregnancy	Wait a year, with 6 months of stable graft function	24, 26

Sharma P et al. AASLD/AST non-graft complications guideline. Liver Transpl 2026;32(8):1157-93 [PMID: 41405235]

Take-home points

- Refer after any decompensating event. Without one, a low MELD isn't a reason to wait if the score misses how sick the patient is
- Acute liver failure or ACLF: call the same day
- Recent drinking calls for assessment and treatment, not an automatic no
- Colorectal metastases and selected cholangiocarcinoma are worth a call
- After transplant, abnormal liver tests mean a call to transplant hepatology the same day
- Check every new drug against tacrolimus



AASLD/AST guidelines: candidate evaluation [PMID: 41405234], graft complications [PMID: 40844852], non-graft complications [PMID: 41405235]

Questions?



Who can get a transplant for intrahepatic CCA?

	With cirrhosis	Without cirrhosis
Tumor	Solitary, unresectable, <3 cm	Unresectable
Biopsy	Required	Required
Before listing	6 months stable after treatment	≥6 months stable on systemic therapy
Outside the liver	No disease	No disease
Setting	Exception points by review (MMaT-3)	Research protocols only

Kodali S, Kulik L, D'Allessio A, et al. ILTS-ILCA consensus. Liver Transpl 2025;31(6):815-31 [PMID: 40014003]. Dove L et al. Hepatology 2026;83(6):1609-45 [PMID: 41405234], statement 11. OPTN transplant oncology review guidance

Appendix: sources (1)

- AASLD/AST candidate evaluation guideline. Hepatology 2026 [PMID: 41405234]
- AASLD/AST graft-related complications guideline. Liver Transpl 2026 [PMID: 40844852]
- AASLD/AST non-graft complications guideline. Liver Transpl 2026 [PMID: 41405235]
- EASL CPG on liver transplantation. J Hepatol 2024 [PMID: 39487043]
- ACG alcohol-associated liver disease guideline. Am J Gastroenterol 2024 [PMID: 38174913]
- ACG acute liver failure guideline. Am J Gastroenterol 2023 [PMID: 37377263]
- AASLD ACLF and critically ill cirrhosis guidance. Hepatology 2024 [PMID: 37939273]
- Lai JC et al. Frailty and sarcopenia guidance. Hepatology 2021 [PMID: 34233031]
- Kodali S et al. ILTS-ILCA iCCA consensus. Liver Transpl 2025 [PMID: 40014003]
- OPTN Policy 9 (effective 10/1/2026) and transplant oncology review guidance
- OPTN/SRTR Annual Data Reports, Liver, 2020-2024 [PMIDs: 35266621, 37132348, 38431359, 39947804, 42276697]

Appendix: sources (2)

- Zarrinpar A, Busuttil RW. Nat Rev Gastroenterol Hepatol 2013 [PMID: 23752825]
- Starzl TE et al. Homotransplantation of the liver in humans. Surg Gynecol Obstet 1963 [PMID: 14100514]
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Appendix: sources (3)

- Louvet A et al. QuickTrans. Lancet Gastroenterol Hepatol 2022 [PMID: 35202597]
- Gustot T et al. CHANCE. J Hepatol 2026 [PMID: 42822699]
- Adam R et al. TransMet. Lancet 2024 [PMID: 39306468]
- Chávez-Villa M et al. LT for colorectal metastases in the US. Liver Transpl 2026 [PMID: 41086433]
- Giorgakis E et al. DCD and machine perfusion in the US. Transpl Int 2026 [PMID: 42294094]
- Seehofer D et al. Non-anastomotic strictures. Am J Transplant 2013 [PMID: 23331505]
- Schneeberger S. Nature 2018 [PMID: 29713063]; J Clin Exp Hepatol 2024 [PMID: 38274508]
- Nasralla D et al. Nature 2018 [PMID: 29670285]; Markmann JF et al. PROTECT. JAMA Surg 2022 [PMID: 34985503]
- van Rijn R et al. DHOPE-DCD. N Engl J Med 2021 [PMID: 33626248]
- Kohli DR et al. ASGE post-LT biliary strictures. Gastrointest Endosc 2023 [PMID: 36797162]
- Tacrolimus (Prograf) prescribing information
- The New York Times, March 5, 2017 (Starzl obituary)